

Methodology and vibrational analysis for measurements on a VTOL RPAS



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Methodology and vibrational analysis for measurements on a VTOL RPAS

Master Thesis in Structure Analysis
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Abstract

In this thesis a methodology for measuring vibrations has been produced and investigated for APID 60, a rotorcraft in a Vertical Take-off and landing remotely piloted aircraft system (VTOL RPAS). A comparative study was carried out for the purpose of identifying the methodology with respect to design modifications common to the APID 60. The pilot-study identified experimental modal analysis (EMA) as a feasible part of the methodology for experimentally extracting the modal parameters of a structure. The EMA was performed on the main frame of the APID 60 where an impact hammer test was chosen as the technique for extracting the response data. As a comparison a point mass was added to the structure to alter the dynamic properties and the test was repeated.

The results from the EMA was compared with a modal analysis performed numerically with a calculation software. Comparison of the results from EMA with the modal analysis performed numerically indicates consistency. This confirms a good reliability of the methodology produced. However, the structure on which the test were preformed is simple in terms of constant structural properties. Further work should therefore investigate whether this methodology of measuring vibrations could be successfully applied to a structure with higher complexity.

keywords:

Experimental modal analysis; impact hammer test; methodology; structural vibration; RPAS; VTOL; Certification.

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Nomenclature

Abbreviations

ADC	Analog digital converter
AMC	Acceptable means of compliance
CS	Certification specifications
DAQ	Data acquisition
DFT	Discrete fourier transformation
dof	Degrees of freedom
EASA	European aviation safety agency
EMA	Experimental modal analysis
FBD	Free body diagram
FEM	Finite element method
FFT	Fast Fourier transformation
FRF	Frequency response function
ICAO	International civil aviation organization
ICP	Integrated circuit-piezoelectric
IEPE	Integrated electronics piezoelectric
JARUS	Joint Authorities for Rulemaking of Unmanned Systems
LURS	Light unmanned rotorcraft systems
mdof	Multiple degrees of freedom
MEMS	Micro electro mechanical systems
PZT	Lead zirconate titanate
RPAS	Remotely piloted aircraft system
RPM	Revolutions per minutes
RPS	Remote pilot station
sdof	Single degree of freedom
tdof	Two degrees of freedom
UAS	Unmanned aircraft systems
UAV	Unmanned aerial vehicle
VTOL	Vertical take-off and landing

Variables

$\bar{F}$	Force vector
$\bar{a}$	Acceleration vector
m	Mass of the particle
k	Spring stiffness
c	Damping coefficient
x	Position coordinate
$\dot{x}$	Velocity of the particle
$\ddot{x}$	Acceleration of the particle
ζ	Damping factor
ω_n	Natural frequency
c_{cr}	Critical damping
F	Sinusoidal force
s	Laplace variable
H	Transfer function
j	Imaginary operator
ω	Angular velocity of the applied force
ω_r	Resonance frequency
$\mathbf{M}$	Mass matrix
$\mathbf{C}$	Damping matrix
$\mathbf{K}$	Stiffness matrix

Chapter 1

Introduction

1.1 Background

CybAero AB with residence in Linköping, Sweden, develops and produces customer specified Vertical Take-Off and Landing Remotely Piloted Aircraft Systems, so called VTOL RPAS, for both civil and defence applications. This system consist mainly of four subsystems, the aircraft, a payload connected to the aircraft, a ground control station and a payload control station.

Priorities in the area of vibration testing has been made and is under development. Therefore this thesis was coordinated as a part to examine the potential of vibration related testing on the aircraft APID60, see figure 1.1.

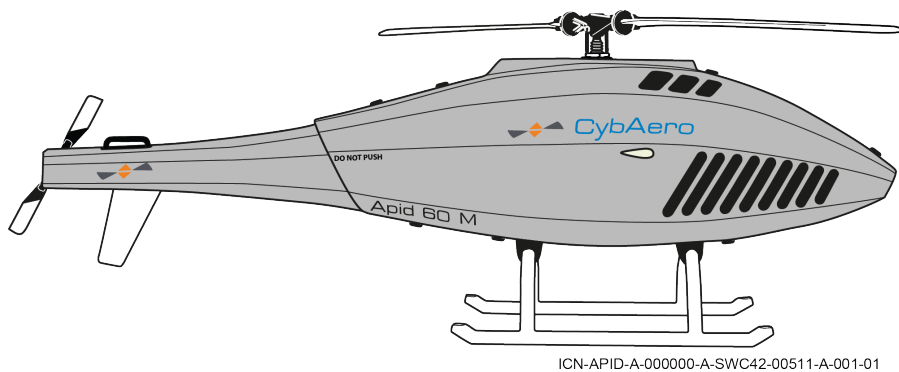


Figure 1.1: The rotorcraft, APID60, of the system.

Helicopters as we know them today are the result of hundreds of failed projects

where some of them contributed with the significant knowledge needed to ultimately construct a successful machine. As J. Gordon Leishman, 2006, put it:

The inherent mechanical and aerodynamic complexities in building a practical helicopter that had adequate power and control and did not vibrate itself to pieces, resisted many ambitious efforts.

A helicopter experience vibrations primarily from the powerplant, transmission, main rotor system, tail rotor system and aerodynamic effects as well as from inherent mechanical imbalances in flight. It is especially important to make sure serious vibrations do not appear in an aircraft with such a high number of flight critical parts as for a helicopter.

The market of light unmanned rotorcraft systems is growing quickly [11], as acknowledged by the International Civil Aviation Organization (ICAO), and authorities struggle to keep up with the increasing amount of aircraft, both commercial and private. The process of certification and regulation is therefore under development.

This thesis is intended as a pilot project where the possibilities for vibration testing in relation to CybAeros needs both now and in the future are investigated. This thesis will suggest a methodology with an associated procedure for the purpose of performing a test where the added value of this type of testing can be examined, evaluated and put into relation to present and future demands both from a construction point of view and a certification point of view.

1.2 Objectives

The objective of this thesis is to produce a methodology suitable for measuring structural vibrations and performing analysis on the rotorcraft APID60 and also exemplify testing by performing actual measurements.

1.2.1 Scope

The objective includes the steps from obtaining equipment to the analysis of data and building a base of theory and facts upon which decisions and conclusions rely. A comparative study is to be performed to better evaluate the methodology.

The steps included in obtaining equipment are:

1. Making a survey of suitable transducers and data acquisition for performing vibration analysis
2. Specifying transducer and data acquisition requirements with respect to intended testing
3. Finding retailers and making a comparison of the equipment
4. Choosing equipment

The steps included in the data acquisition and analysis are:

1. Making a survey of suitable tests
2. Specifying the purpose of the test
3. Creating a test plan with a test procedure
4. Perform testing for the comparative study
5. Analyse data

The methodology is then evaluated with respect to the results obtained from the testing and assessed utility in future work.

1.3 Limitations

The main focus in this thesis lies in producing a methodology. Therefore, less care has been given to provide optimal prerequisites for the tests regarding evaluation of the methodology.

Optimal prerequisites here includes and refers to measuring equipment, test set up, collection of experimental data and data analysis. To keep the report within the specified time frame of the thesis one sub assembly of the APID60 will be under investigation.

1.4 Method

The steps involved in reaching the objective includes;

- Building a theoretical background
- Finding the required specifications of equipment

- Finding equipment
- Choosing appropriate test methods
- Constructing test plan with detailed procedure
- Performing test
- Make test report
- Evaluate and validate the results from the test with computer simulation
- Evaluate the methodology

By keeping the latest certification requirements on light unmanned rotorcraft systems as a basis when designing and evaluating a methodology for vibration testing increased utility of the methodology can be ensured for future tests.

Vibration testing forms a natural bridge between experimental tests and calculation, therefore this thesis will involve the calculation department of CybAero to evaluate results from the test by simulations in CAD environment.

1.5 Test Object APID 60

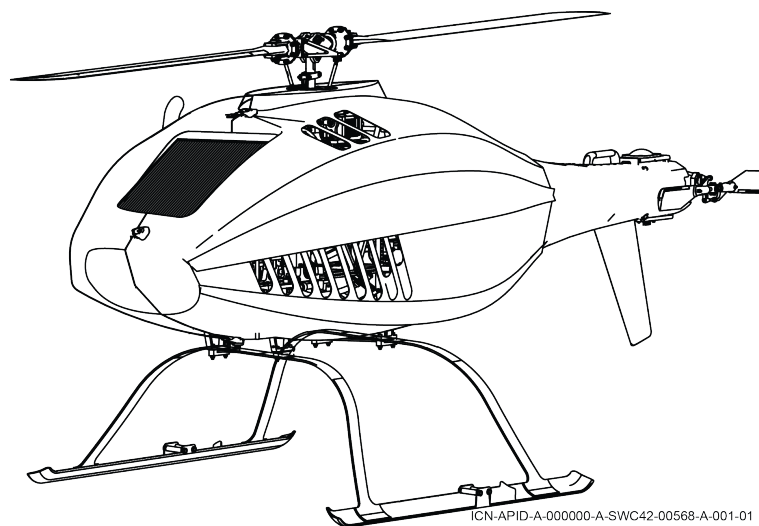


Figure 1.2: APID60

The APID60 is a Vertical Take-Off and Landing Unmanned Aerial Vehicle (VTOL UAV) as a part of a Unmanned Aerial System (UAS) where it operates as a mobile payload-carrying platform. The payload is typically a camera but can be exchanged depending the type of mission.

Specifications:

Max. Weight:	220 Kg
Length:	3.20 m
Height:	1.30 m
Width:	1.20 m

Through discussions a simplification of the helicopter was made in terms of sub-structures where the APID60 was divided in three different parts and seen as independent and interconnected stiff bodies. Namely, the main frame, engine frame and the tail boom. After further reasoning it was decided that tests on the main frame would be sufficient. The main frame is shown in figure 1.3.



Figure 1.3: The test object and the main frame of APID60

1.6 Thesis Outline

Chapter 2 describes the current certification specification and regulations that are related to vibration testing, taken from the document *JARUS CS-LURS V.1.0*.

Chapter 3 provides the necessary theoretical knowledge in order to understand and analyze the results obtained from the measurements. It explains by starting

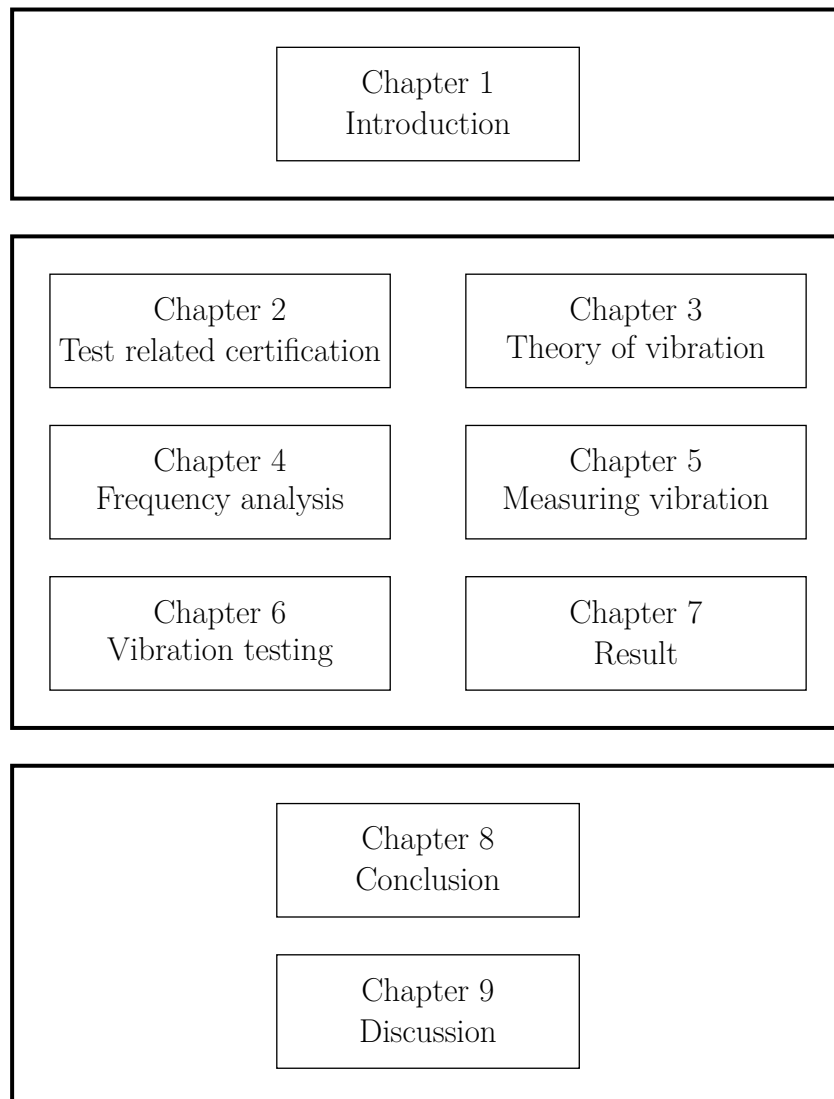


Figure 1.4: The outline of the thesis

with a basic single degree of freedom and continues with a multiple degree of freedom system. The phenomena of resonance and the meaning of a transfer function for a forced vibration is hereby explained.

Chapter 4 contains methods for frequency analysis. Fast Fourier Transformation and window functions are covered.

Chapter 5 covers the main components in the chain of vibration analysis. Different types of transducers are examined and compared along with factors that

distorts the retrieved signal from the vibration.

Chapter 6 presents the method chosen for vibration testing and the different test activities related to the testing.

Chapter 7 presents the results of the thesis. Here the test method, equipment, software and data from the vibration analysis is stated.

Chapter 8 and **Chapter 9** contains the conclusion drawn from the study as well as recommendations for future work and discussion around the topic of the report. Schematics of the thesis outline is displayed in figure 1.4.

Chapter 2

Test Related Certification

2.1 Background to Chapter

This chapter is a part of the pilot study and is devoted to tests concerning vibrations and related to the process of certification. By looking at official documents specifying tests for light unmanned rotorcraft systems one can gain knowledge about relevant tests ought to be performed, and thus what kind of equipment used for the tests can be identified. This thesis focuses on vibration testing, so tests that are not regarding vibrations will be omitted.

2.2 Introduction

International Civil Aviation Organization (ICAO) develops fundamental standards and recommended practises for integration of UAS into airspace and was established by the convention of international civil aviation, known as Chicago convention, and was signed by a total of 191 member states as of 2013 [11] [15]. In Annex 8 of the Chicago convention the standards of certification are given and applies to unmanned aircraft as well [11].

The primarily purpose of certification is to produce safe aircrafts that will allow for safe operations. Annex 8 has recognized three products so far that require airworthiness approval [11]:

1. Aircraft
2. Engines
3. Propellers

A RPAS however, uses a Remote Pilot Station (RPS) and this has not yet been introduced in Annex 8, but is about to be introduced as a new aeronautical product that can be certified separately [11].

Established in 2002 the European Aviation Safety Agency (EASA) is the European Union authority in aviation safety and coordinates safety management, handles the certification of aviation products and has oversight on approved organisations and EU member states [13].

EASA is mandated by regulation (EC) No 216/2008 to regulate both UAS and RPAS for an operating mass above 150 Kg for civil applications [16]. EASA is also a member of Joint Authorities for Rulemaking of Unmanned Systems, so called *JARUS*. JARUS consists of experts from the National Aviation Authorities (NAA) and was put together to develop proposals for all aspects of UAS regulation [14]. Among them are Certification Specifications for Light Unmanned Rotorcraft Systems (CS LURS) below 600 Kg [16] and their intentions are to contribute to other rulemaking efforts both regional and worldwide [14].

The by JARUS produced document *JARUS CS-LURS V.1.0* will be used as a base when considering equipment and methodologies for testing on the APID 60. Relevant parts of the document are covered in section 2.3.

JARUS CS-LURS V.1.0 is an official document, 128 pages long, that covers certification specification of light unmanned rotorcraft systems with a weight below 600 kg. It is structured in two parts, *Book 1 Airworthiness code* and *Book 2 Acceptable means of compliance (AMC)* where each part is subdivided in parts that covers a specific topic.

Since APID 60 is within the range of the above specified regulation it will have to meet the requirements dictated by these authorities in order to be certified. By identifying parts of the document that regards vibration measurements or analysis an understanding of what kind of measuring equipment and type of tests that needs to be acquainted with can be identified. This will ensure that maximum utility of the equipment will be provided for.

2.3 CS LURS

The following list contains sections from the document *JARUS CS-LURS V.1.0* that was identified as relevant from a structural dynamic point of view.

- CS-LURS.241 Ground resonance, page 12
- CS-LURS.629 Flutter, page 27
- CS-LURS.663 Ground resonance prevention means, page 28
- CS-LURS.931 Shafting critical speed, page 37
- AMC LURS.571(a)(3) Fatigue evaluation of flight structure, page 112
- AMC LURS.907 Vibration, page 116
- AMC B-LURS.33 Vibration, page 125

2.4 Summary

This section summarizes points from the document *JARUS CS-LURS V.1.0* that indicate utilization of vibration measuring equipment.

2.4.1 Certification Specification

The absence of ground resonance occurrence must be demonstrated. Several options are given in order to satisfy this requirement. It is stated that determination of the probable range of vibrations during service must be established. This can be done by measuring vibrations on the helicopter during a run-up and a run-down cycle of the engine while the helicopter stands firmly on the ground.

Flutter on aerodynamic surfaces must be absent during operation under different speeds and power conditions. Determining vibrational reference values through measurements where amplitude are measured given a specific range of speed and power. The results can be compared and conclusions regarding flutter can be made.

The specification of shaft critical speeds suggests that, by demonstration, the critical speeds of any shafting must be determined. If the critical speed lies within proximity for idling, power-on or auto-rotative conditions the stress must be shown by tests to be adequately low. The critical speed represents the angular velocity of the shaft that excites the resonance frequency of the shaft, so it is thus the resonance frequency translated into angular velocity of the shaft. Finding these frequencies can be done in a test rig where the amplitude is measured as a function of angular velocity.

2.4.2 Acceptable Means of Compliance

The fatigue evaluation of the flight structure through analysis should be validated with local in-flight measurements data. Acceleration is one of three options explicitly expressed in the text. This part strongly emphasise the utility of vibration measurement equipment in the process of certification.

In order to determine that there are no harmful vibrations in the rotor drive system performing a vibration survey is suggested. The vibrations survey will determine the natural frequencies for the system components and the frequencies produced by the engine and the goal is to tune the system to damp out vibration peaks and if necessary shift resonances away from the engine RPM range. The acquisition of this vibration data relies on vibrations measurement equipment.

Chapter 3

Theory of Vibrations

3.1 Background to Chapter

The purpose of this chapter is to provide basic fundamental theory behind structural vibrations. A vibration can be seen as an oscillation of a body about its equilibrium position where the mass and a restoring force is interacting to produce the motion. Vibrations can be put into four categories [4], namely:

1. Undamped free vibrations
2. Undamped forced vibrations
3. Damped free vibrations
4. Damped forced vibrations

In reality damping is always present, i.e. energy is dissipated from the system due to the motion. This means in practice that only *damped free vibrations* and *damped forced vibrations* exists.

The sections below will cover both *single-degree-of-freedom (sdof)*, and *two-degree-of-freedom (tdof)*, systems for damped free vibrations and damped forced vibrations respectively. The sdof will demonstrate the effects of damping on the behaviour of the system whereas the mdof system will be used to analyse the dynamic properties. This is important to understand in order to follow the analysis of data from measurements.

In this thesis a sdof system refers to a system that can be described solely by a single position coordinate and hence a tdof system refers to a system described by two positions coordinates.

3.2 Damped Free Vibrations

The sdof system depicted in figure 3.2 will be analysed for the purpose of cover the theoretical background needed for understanding results obtained in vibrational testing. By doing this the fundamental behaviour of a vibrating system can be observed. In this simple system a mass is coupled to the ground with a spring and a viscous damper. This mass will vibrate freely since no external force is applied. However, vibration will be present after introducing an initial velocity.

Through Newtons second law, equation 3.1, the equations of motion can be derived for the particle of a damped free vibration depicted in figure 3.1a.

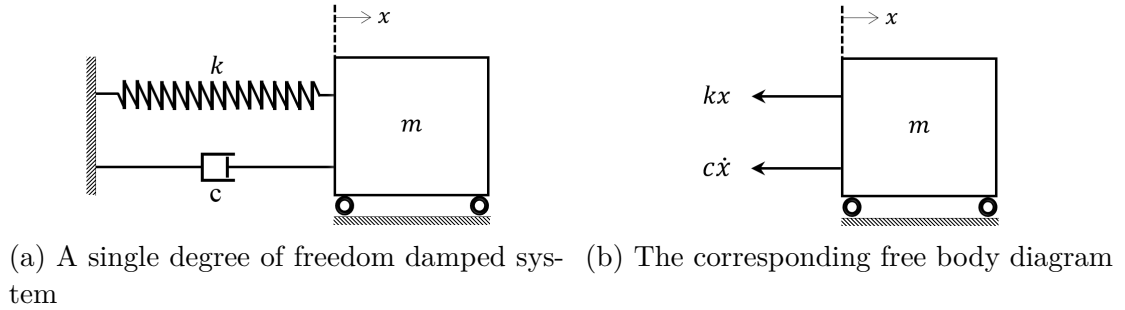


Figure 3.1: Sdof system

$$\sum \bar{F} = m\bar{a} \quad (3.1)$$

where

$\bar{F}$ is the force vector

$\bar{a}$ is the acceleration vector

m is the mass

The resulting equation of motion from figure 3.1b yields a second order differential equation

$$\ddot{x} + \frac{c}{m}\dot{x} + \frac{k}{m}x = 0 \quad (3.2)$$

where

k is the spring stiffness

c is the damping coefficient

m is the mass of the particle

x is the position coordinate

$\dot{x}$ is the velocity of the particle

$\ddot{x}$ is the acceleration of the particle

The following definitions are used to simplify equation 3.2.

Natural frequency

$$\omega_n = \sqrt{\frac{k}{m}} \quad (3.3a)$$

Critical damping

$$c_{cr} = 2m\omega_n \quad (3.3b)$$

Damping factor

$$\zeta = \frac{c}{c_{cr}} = \frac{c}{2m\omega_n} \quad (3.3c)$$

By using definitions 3.3a - 3.3c equation 3.2 can be rewritten as

$$\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2x = 0 \quad (3.4)$$

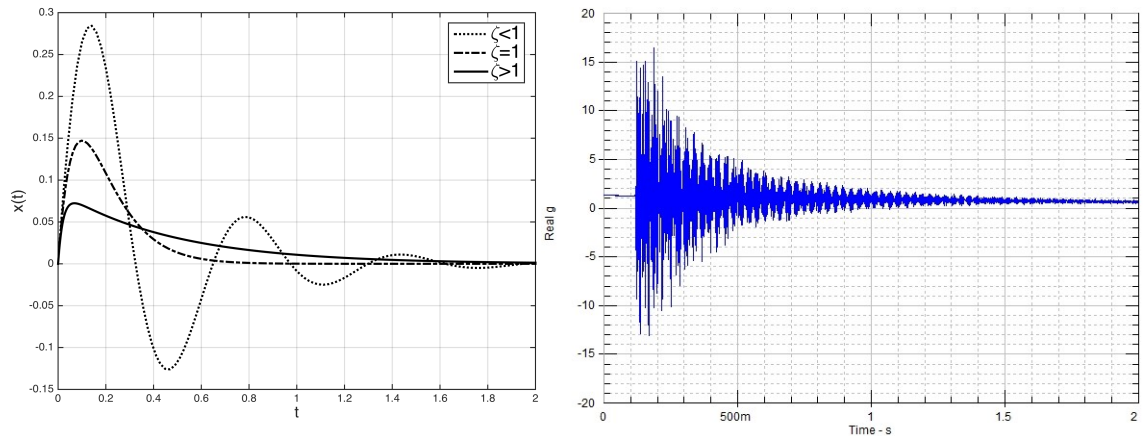
as known from textbooks [4]. The solution to the differential equation 3.4 depends strongly on the damping factor ζ and are therefore divided in three categories, namely:

$\zeta < 1$ under damped

$\zeta = 1$ critically damped

$\zeta > 1$ over damped

Solving the differential equation for the three cases and plotting the result yields figure 3.2a. In lightweight metal structures the case of under damping is prominent, see figure 3.2b.

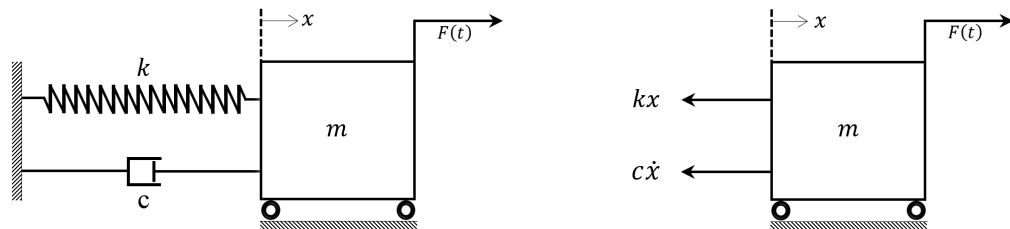


(a) The solution of equation 3.4 for three cases of damping. (b) Under damped response of an aluminum structure

Figure 3.2

3.3 Damped Forced Vibrations

A damped forced sdof system is good for introducing the idea of viewing the system as a black box system with fixed system characteristics. In this case an external sinusoidal force is applied to the single mass system, figure 3.3a. The equation of motion is now represented by equation 3.5.



(a) A single degree of freedom damped system with forced excitation (b) The corresponding free body diagram

Figure 3.3: Sdof system with an applied external sinusoidal force

$$\ddot{x} + \frac{c}{m}\dot{x} + \frac{k}{m}x = \frac{F}{m} \quad (3.5)$$

where

F is a sinusoidal force varying with time

Now that the system is excited by an external force varying with time rather than an impulse it is interesting to analyse the response for different frequencies. Section 3.3.1 is devoted to that.

3.3.1 Transfer Function

The sdof system can be viewed as black box where an input creates an output that depends on the unknown characteristics of the box, see figure 3.4.

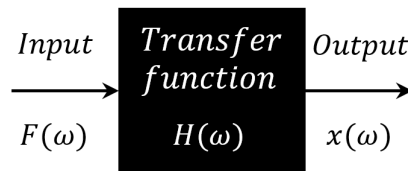


Figure 3.4: Black box model of the system

In order to get the transfer function of equation 3.5 the Laplace transform has to be found. The Laplace transform of 3.5 with initial conditions set to zero is

$$s^2x(s) + \frac{c}{m}sx(s) + \frac{k}{m}x(s) = \frac{F(s)}{m} \quad (3.6)$$

where

s is the Laplace variable

The transfer function is the ratio between the position coordinate and the external force

$$\frac{x(s)}{F(s)} = \frac{1/m}{s^2 + \frac{c}{m}s + \frac{k}{m}} \quad (3.7)$$

Using the definitions 3.3a to 3.3c to rewrite as before

$$\frac{x(s)}{F(s)} = \frac{1/m}{s^2 + 2\zeta\omega_n s + \omega_n^2} = H(s) \quad (3.8)$$

where

$H(s)$ is the transfer function

3.3.2 Resonance Frequency

To calculate the frequency response s is replaced by $j\omega$ according to equation 3.9

where

j is the imaginary operator

ω is the angular velocity of the applied force

$$s = j\omega \quad (3.9)$$

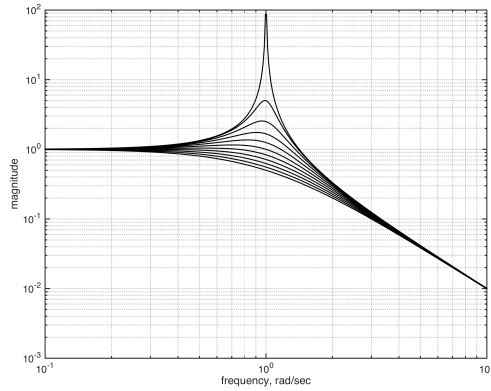
Putting equation 3.9 into equation 3.8 yields the following complex expression

$$\frac{x(j\omega)}{F(j\omega)} = \frac{1/m}{-\omega^2 + 2\zeta\omega_n j\omega + \omega_n^2} \quad (3.10)$$

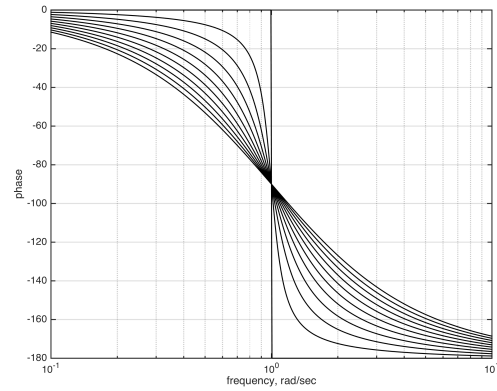
The magnitude $|\frac{x(j\omega)}{F(j\omega)}|$ and phase $\angle \frac{x(j\omega)}{F(j\omega)}$ of 3.10 is plotted versus frequency in figure 3.5.

By looking closely on the magnitude plot, figure 3.5a, one can see that for $\zeta > 0$ the maximum amplitude occurs at frequencies slightly lower than the natural frequency. The explanation is that the resonance frequency, ω_r is the frequency for which the magnitude value of the ratio between output and input is maximum.

The resonance frequency is hence calculated by finding the maximum value of $|\frac{x(j\omega)}{F(j\omega)}|$.



(a) Magnitude plot



(b) Phase plot

Figure 3.5: Magnitude and phase are plotted for the complex transfer function with respect to frequency for $\zeta = [0.1, 0.2, \dots, 1]$ and $\zeta \approx 0$

The solution sought is described in equation 3.11, also known from textbooks [4].

$$\omega_r = \omega_n \sqrt{1 - 2\zeta^2} \quad (3.11)$$

As long as there is damping present in a material, this is the frequency for which a sdof system would have its maximum vibrating amplitude, not the natural frequency.

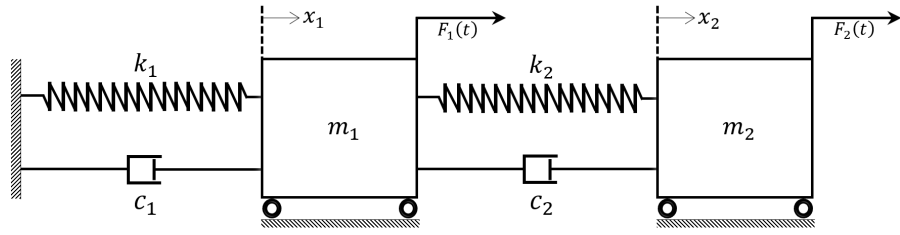
3.4 Two Degrees of Freedom

Complexity quickly increases as more degrees of freedom are introduced. For this reason a two degree of freedom system is chosen to show the characteristics. Consider the system in figure 3.6 where mass m_2 is added and coupled with a spring and a damper to mass m_1 .

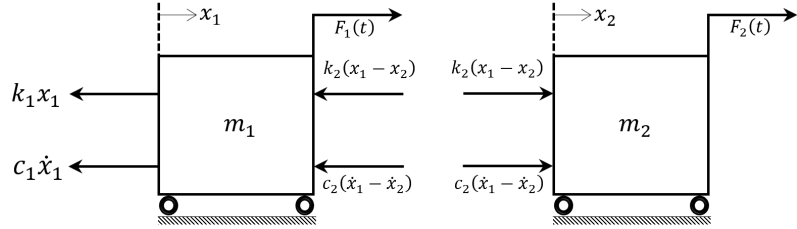
We get two equations of motion, one for each degree of freedom.

$$m_1 \ddot{x}_1 + k_1 x_1 + c_1 \dot{x}_1 + k_2(x_1 - x_2) + c_2(\dot{x}_1 - \dot{x}_2) = F_1(t) \quad (3.12a)$$

$$m_2 \ddot{x}_2 - k_2(x_1 - x_2) - c_2(\dot{x}_1 - \dot{x}_2) = F_2(t) \quad (3.12b)$$



(a) Dual mass two degree of freedom system



(b) The corresponding FBD

Figure 3.6: Tdof system

Written on the matrix form

$$\mathbf{M}\ddot{\mathbf{x}} + \mathbf{C}\dot{\mathbf{x}} + \mathbf{K}\mathbf{x} = \mathbf{F} \quad (3.13a)$$

$$\begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix} \begin{bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{bmatrix} + \begin{bmatrix} c_1 + c_2 & -c_2 \\ -c_2 & c_2 \end{bmatrix} \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} + \begin{bmatrix} k_1 + k_2 & -k_2 \\ -k_2 & k_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} F_1(t) \\ F_2(t) \end{bmatrix} \quad (3.13b)$$

where

$\mathbf{M}$ is the mass matrix

$\mathbf{C}$ is the damping matrix

$\mathbf{K}$ is the stiffness matrix

Taking the Laplace of equation 3.13b yields

$$\begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix} \begin{bmatrix} s^2 x_1 \\ s^2 x_2 \end{bmatrix} + \begin{bmatrix} c_1 + c_2 & -c_2 \\ -c_2 & c_2 \end{bmatrix} \begin{bmatrix} s x_1 \\ s x_2 \end{bmatrix} + \begin{bmatrix} k_1 + k_2 & -k_2 \\ -k_2 & k_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} F_1(s) \\ F_2(s) \end{bmatrix} \quad (3.14)$$

Rearranging as

$$\mathbf{A}\mathbf{x} = \mathbf{F} \quad (3.15a)$$

$$\begin{bmatrix} s^2m_1 + s(c_1 + c_2) + (k_1 + k_2) & -sc_2 - k_2 \\ -sc_2 - k_2 & s^2m_2 + sc_2 + k_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} F_1(s) \\ F_2(s) \end{bmatrix} \quad (3.15b)$$

Solving for $\mathbf{x}$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{1}{\det(\mathbf{A})} \begin{bmatrix} s^2m_2 + sc_2 + k_2 & sc_2 + k_2 \\ sc_2 + k_2 & s^2m_1 + s(c_1 + c_2) + (k_1 + k_2) \end{bmatrix} \begin{bmatrix} F_1(s) \\ F_2(s) \end{bmatrix} \quad (3.16)$$

where

$$\det(\mathbf{A}) = [s^2m_1 + s(c_1 + c_2) + (k_1 + k_2)][s^2m_2 + sc_2 + k_2] - [-sc_2 - k_2][-sc_2 - k_2] \quad (3.17)$$

and the transfer function matrix is

$$\mathbf{H} = \begin{bmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{bmatrix} = \frac{1}{\det(\mathbf{A})} \begin{bmatrix} s^2m_2 + sc_2 + k_2 & sc_2 + k_2 \\ sc_2 + k_2 & s^2m_1 + s(c_1 + c_2) + (k_1 + k_2) \end{bmatrix} \quad (3.18)$$

The number of transfer functions for tdof is thus 4. For dof higher than 1 the number of possible transfer functions will be equal to the number of dof squared. However, the transfer function matrix $\mathbf{H}$ is symmetrical [6], equation 3.19

$$\mathbf{H}^T = \mathbf{H} \quad (3.19)$$

As can be seen from equation 3.18 three unique transfer functions exist. The frequency domain representation of 3.18 see figure 3.7.

The symmetry of the transfer function matrix suggests that Maxwell's principle of reciprocity applies, see equation 3.20 [6].

$$\mathbf{H}_{ij} = \mathbf{H}_{ji} \quad (3.20)$$

where

i represents the response

j represents the input

This is of major importance when performing experimental tests to obtain the modal characteristics of a structure since this suggests an admissible set of transfer functions for the structure. This will be covered further in subsection 3.4.1. In figure 3.7 the typical shape of a tdof transfer function can be seen along with the phase. Two distinct peaks, resonances, as well as a dip, antiresonance, can be distinguished.

3.4.1 Maxwell's Reciprocal Theorem

As mentioned in section 3.4 Maxwell's reciprocal theorem holds for the transfer function matrix.

It states that the response H_{ij} at section i due to a force applied at section j is equal to the response H_{ji} at section j due to a force applied at section i , if the excitations are identical [7] [6], see equations 3.21.

For an identical excitation

$$F_i = F_j \quad (3.21a)$$

due to the symmetry of the transfer function, the response on two different locations are identical

$$X_i = X_j \quad (3.21b)$$

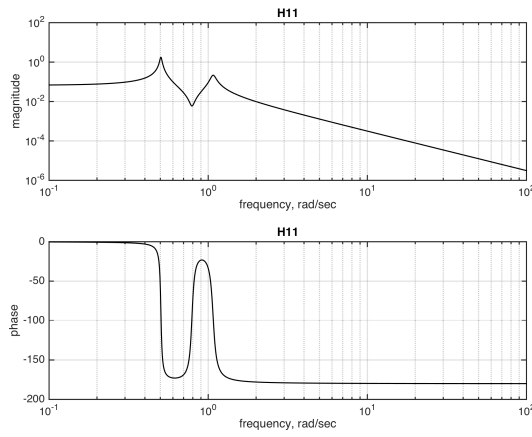
where

$$X_i = H_{ij}F_j \quad (3.21c)$$

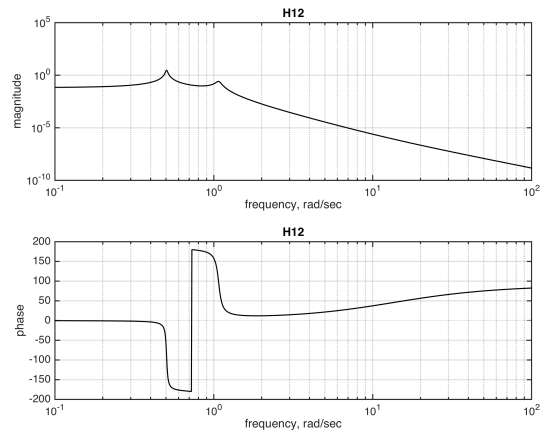
and

$$X_j = H_{ji}F_i \quad (3.21d)$$

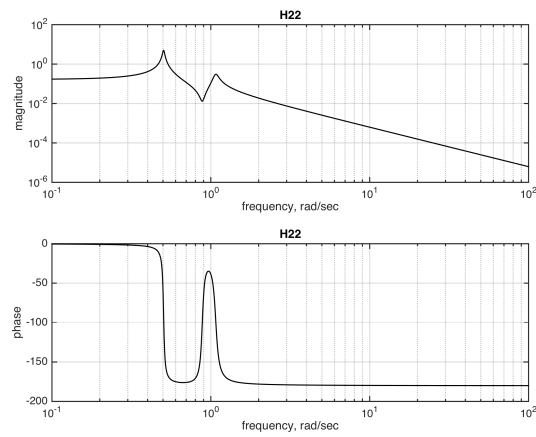
This means in practice that the same transfer function will be extracted after swapping place of the excitation location and the response location, given that the material is linear.



(a) Transfer function H_{11}



(b) Transfer function H_{12}



(c) Transfer function H_{22}

Figure 3.7: Unique transfer functions for the tdof system

Chapter 4

Frequency Analysis

4.1 Background to Chapter

This chapter will cover the main form of frequency analysis, namely fast Fourier transformation, by demonstrating the use in practice. The importance of using a proper window for the captured time signal will be explained as well.

4.2 Fast Fourier Transformation

Measured signals are represented in the time domain where the signal is given by amplitude and time. This representation does not reveal all information interesting from an analysis point of view. By transferring the signal to the frequency domain the spectrum can be analysed.

An algorithm called fast Fourier transformation (FFT) is often used for this purpose [5]. To demonstrate the use of FFT the following five functions are considered

$$x_1 = A_1 \sin(2\pi f_1 t) \quad (4.1a)$$

$$x_2 = A_2 \sin(2\pi f_2 t) \quad (4.1b)$$

$$x_3 = A_3 \sin(2\pi f_3 t) \quad (4.1c)$$

$$x_4 = A_4 \sin(2\pi f_4 t) \quad (4.1d)$$

$$x = x_1 + x_2 + x_3 + x_4 \quad (4.1e)$$

Equation 4.1a-4.1d can represent excitations from different components on a structure where 4.1e is the sum of the vibrations felt by a measuring device.

In figure 4.1a the components of the combined signal equation 4.1e are plotted in the time domain where both the amplitude and the cyclic frequency is explicitly expressed. Figure 4.1b shows the combined signal of the signal components in both time domain and frequency domain. The frequency components and the amplitudes are hard to distinguish by looking at the time domain in figure 4.1b but transferring the data of the combined signal to the frequency domain the amplitudes and frequency content are easily distinguished.

A discrete Fourier transformation (DFT) of the signal x has been performed with a fast Fourier transformation algorithm. As can be seen the FFT displays the frequency components of the combined signal and the relative amplitudes of each component of vibration.

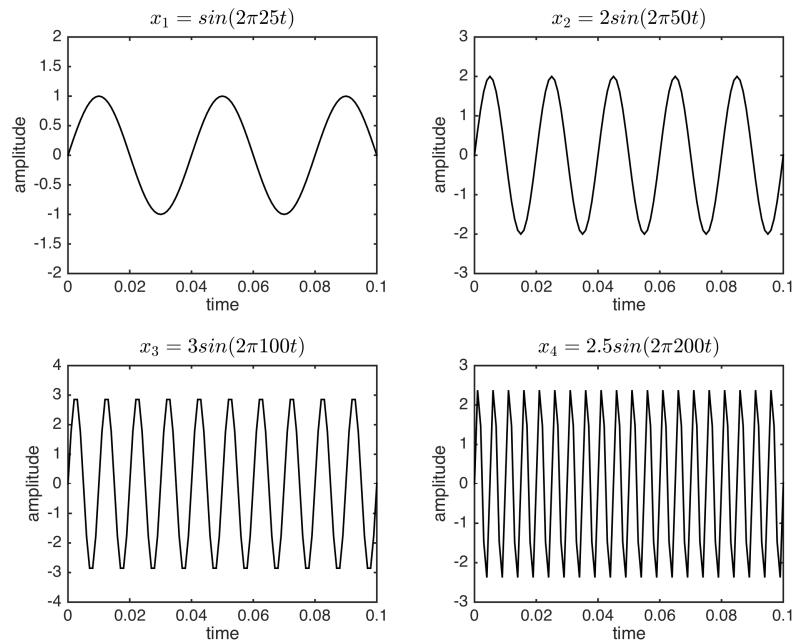
4.2.1 Window Functions

A few words about window functions will be mentioned here as they are important in frequency analysis. A window function is applied on the measured time domain data before the FFT. The purpose of using a window function is to reduce leakage in spectral energy in the frequency domain thus providing more accurate results. Leaking occurs when the time length of the measured segment are not a multiple of the time period of the measured signal [12]. This causes misleading conclusions and problems when analysing the data in the frequency domain since the spectral energy is distributed on frequencies not present in the signal analysed. By applying a correct window function to the signal leaking can be reduced [12].

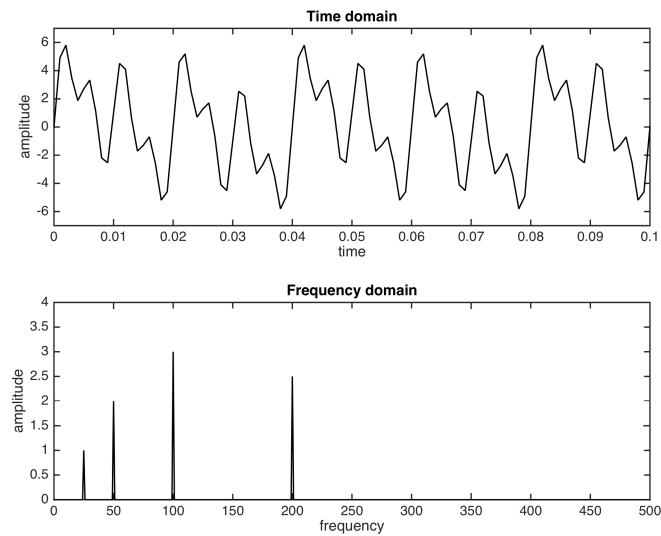
There are several window functions that can be applied to the time domain data [12]. Four of them are presented here, namely: Hanning, Hamming, Gaussian and exponential, see figure 4.2. The windows are multiplied to the time signal to reduce the amplitudes according to the shape of the function. Compare the time domain signals for figure 4.3a and figure 4.3b.

This have the effect of making the signals more periodic and the result is reduced leakage in the succeeding FFT calculation. See the frequency domain representation of the signal in figure 4.3a and figure 4.3b.

This demonstrates the use of windowing where a function $\sin(2\pi 105t)$ without and with windowing is compared. The ideal FFT of this signal would have a single thin peak of amplitude 1 at 105 Hz. The result of the FFT depends on



(a) Time domain representation of the signal components x_1 , x_2 , x_3 , x_4 .



(b) The combined signal x in time domain and the representation of the signal in the frequency domain respectively.

Figure 4.1: The use of FFT

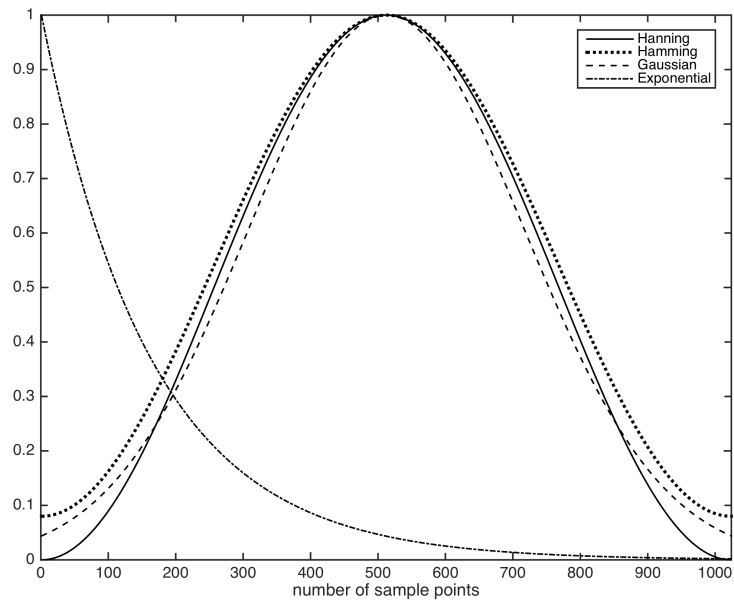
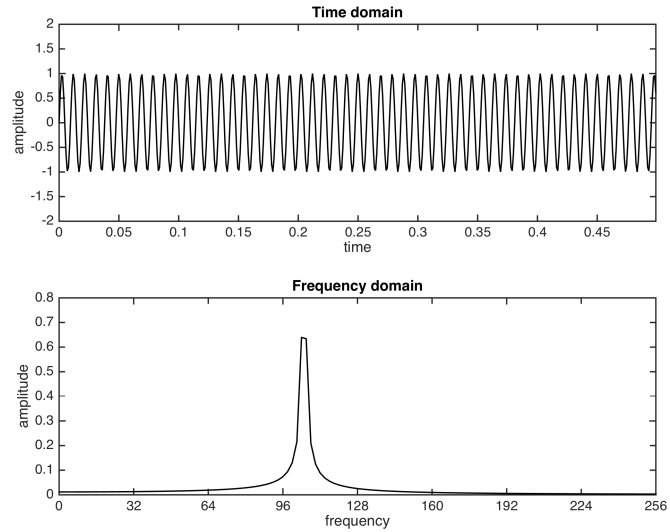
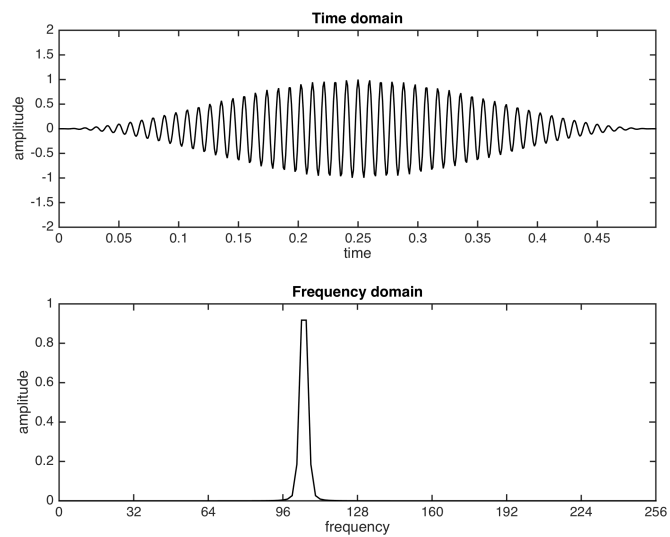


Figure 4.2: Windows

more factors than the window function alone. Data acquisition parameters such as number of data points and sampling rate will affect the overall quality of the measured data. By comparing both FFTs in figure 4.3 the reduction of leakage to other frequencies has been reduced in figure 4.3b, the FFT is not ideal but the result is better.



(a) Time signal without window



(b) Time signal with Hanning window

Figure 4.3: Comparison between the frequency domain of $\sin(2\pi 105t)$ for a cut and a windowed time signal.

Chapter 5

Measuring Vibration

5.1 Background to Chapter

As a part of the pilot study this chapter covers the main components in the chain of vibration analysis. Necessary information about a few selected transducers used for vibration analysis are analysed and compared along with data acquisition equipment.

5.2 Transducers for measuring Vibration

In order to translate the mechanical motion of the vibration into an electrical signal and store the data an electromechanical transducer is needed. There are several types of transducers that measure acceleration. Among them are proximity probes, capacitive probes, piezoresistive and piezoelectric transducers. Two types of accelerometers were considered further, namely: piezoelectric accelerometers and MEMS accelerometers.

5.3 The Piezoelectric Accelerometer

The sensing element and thus the key component of a piezoelectric accelerometers is the piezoelectric ceramic or crystal. The material used for this element can either be man-made or taken from nature. Most commonly the material used is a man-made ferroelectric ceramic. The material has been artificially polarized in a process called *poling* to produce the piezoelectric effect. An example is *Lead Zirconate Titanate*, $Pb(ZrTi)O_3$ or *PZT* for short. *Quartz* is an example of a naturally occurring piezoelectric material that is used. Generally, man-made piezoelectric materials have higher sensitivity due to the artificial poling but in return

has lower thermal and physical stability compared to the naturally occurring materials. For a comparison of the sensitivities between quartz and PZT see table 5.1.

Materials that exhibit the piezoelectric effect will create a charge on the surface of the crystal that is proportional to the strain when a force is acting upon it. They also show the reverse piezoelectric effect, i.e. they will be subjected to a mechanical strain when under influence of an electrical field.

The piezoelectric property can be described both by a sensitivity for charge and voltage.

The sensitivities of the two above mentioned materials are listed in 5.1

Material	Charge sensitivity [pCN^{-1}]	Voltage sensitivity [$mVmN^{-1}$]
PZT	110	10
Quarts	2.5	50

Table 5.1: Sensitivities for PZT and Quarts

Important to notice is that a piezoelectric material is unable to hold a charge due to a statical force. This limits the piezoelectric accelerometer to only measure dynamic events.

The fundamental components of a piezoelectric accelerometer is the piezoelectric element, the seismic mass and the accelerometer base. These components are connected mechanically, where a seismic mass is coupled to a piezoelectric element and the piezoelectric element is then connected to the accelerometer base. The base is mounted to the surface of the object that is vibrating thus leading the movement of the object through the piezoelectric element to the seismic mass. The acceleration of the seismic mass will exert a force upon the element. This force strains the element producing a charge on its surface, from this charge the force can be calculated. For a known seismic mass the acceleration can be calculated. With other words, the seismic mass converts the acceleration input to a stress that creates an output charge signal proportional to the momentary acceleration. The dynamic properties of the accelerometer, however, will affect the output thus limiting the useful frequency range. For a schematic diagram see figure 5.1.

There are three main ways for the seismic mass to strain the piezoelectric element, see figure 5.2.

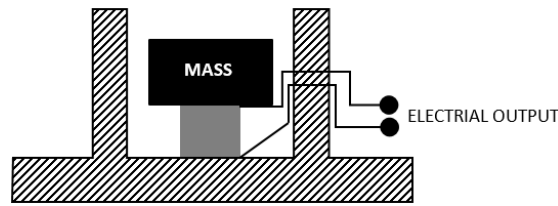


Figure 5.1: Schematic diagram of an accelerometer

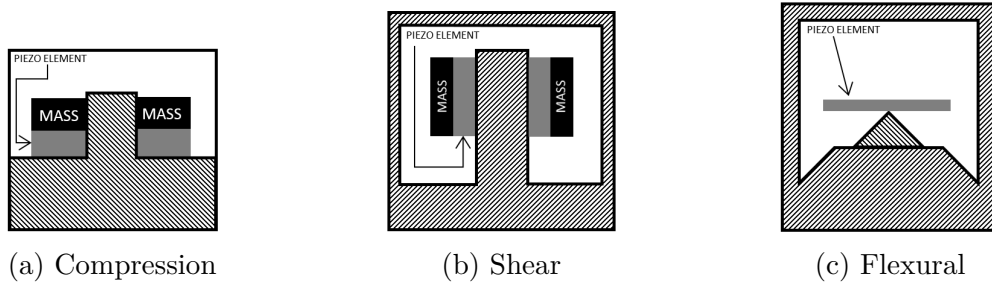


Figure 5.2: Schematic diagrams of different types of loading an accelerometer

5.3.1 Charge mode

Piezoelectric accelerometers can be divided into two groups concerning their output signal, namely charge and voltage accelerometers. Basically, the charge mode accelerometer has no integrated electronics whereas the voltage mode accelerometer has.

A charge mode piezoelectric accelerometer outputs the signal generated on the surface of the crystal directly. This is a high-impedance electrical charge signal. In order to use this signal an external charge amplifier is needed to convert the signal to a low-impedance voltage signal.

Since the signal between the accelerometer and the charge amplifier is of high-impedance nature it is prone to environmental contamination. Low noise coaxial cables needs to be used in order to reduce triboelectric noise effects (noise originating from movement of the cable).

Since the integrated electronics for a charge amplifier is omitted in a charge mode accelerometers it benefits from having a higher temperature operating limit compared to a voltage mode accelerometer. The external charge amplifier needed in the test setup for a charge mode accelerometer provides the option to change settings unlike fixed output characteristics of a integrated charge amplifier.

5.3.2 Voltage mode

The voltage mode piezoelectric accelerometers uses built-in signal conditioners within the housing of the sensor. They are known as *Integrated Electronics Piezoelectric* abbreviated *IEPE* and also denoted by voltage mode. The integrated circuit is a charge amplifier that converts the high-impedance charge signal to a low-impedance voltage signal. Accelerometers from the manufacturer PCB that contain integrated circuits are called ICP® for Integrated Circuit-Piezoelectric. Benefits concerning a voltage mode piezoelectric accelerometer are the use of long standard coaxial cables without an increase in noise or loss of resolution.

A voltage mode piezoelectric accelerometer needs to be supplied with a constant excitation current in order to work. This is in the range of $2mA$ to $20mA$. The integrated circuit renders the accelerometer less durable, gives a lower operational temperature limit, sensitivity for electrostatic discharge and fixed output characteristics.

5.4 The MEMS Accelerometer

MEMS stands for *Micro Electro Mechanical Systems* and have mechanical dimensions below $100\mu m$. They are produced by a technique called micro-fabrication technology rather than conventional fabrication [10].

Common MEMS accelerometers are based on piezoresistive effect or use the change of capacitance to sense acceleration. Other ways to accomplish the sensing are present too, such as the thermal MEMS accelerometers that relies on heated gas molecules to detect acceleration [9]. To stay within the time frame of the work this thesis considered only capacitive MEMS accelerometers for the vibration analysis.

The capacitive MEMS accelerometer senses the change in capacitance as a finger of a spring-suspended seismic mass moves between two other fingers, see figure 5.3. This movement changes the capacitance. By relating the change of capacitance to the equations of motion for the spring-mass system and voltage balance the acceleration can be described as a function of the voltage.

5.5 Impact Hammer

An impact hammer is used to introduce an excitation to a structure. The force of the impact applied by the hammer is measured by a force transducer integrated

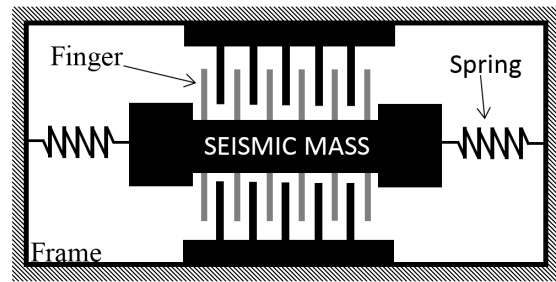


Figure 5.3: Schematic diagram of a capacitive MEMS, top view

close to the impact tip of the hammer. The sensing element of the force transducer is a piezoelectric crystal [12]. A schematic diagram of an impact hammer can be seen in figure 5.4. A typical measured impact of an excitation in the time domain

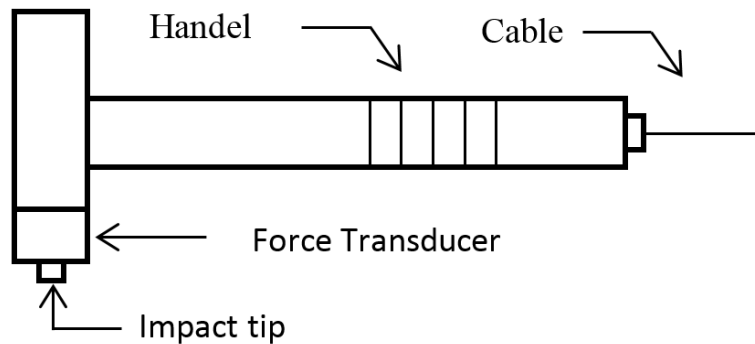


Figure 5.4: Impact hammer

is shown in figure 5.5. Changing the hardness of the impact tip is the only way how to control the frequency band of excitation [12]. A hard impact tip provides a wider frequency band whereas a soft hammer tip gives a narrow band of excitation.

5.6 Data Acquisition

Transducers used for sensing can either output an analog or digital signal. This thesis focuses on accelerometers that has a sensing unit that translates the physical quantities to an analog signal. For this continuously defined signal to be interpreted by a computer it has to be digitized by being sent through a front end or *DAQ* which in this case is an *Analog-Digital converter or ADC*. The ADC will sample the continuous signal at discrete time points and represent them in a digital form. How well the ADC represents the continuous signal depends on several factors including sample rate and the bit depth of each sample.

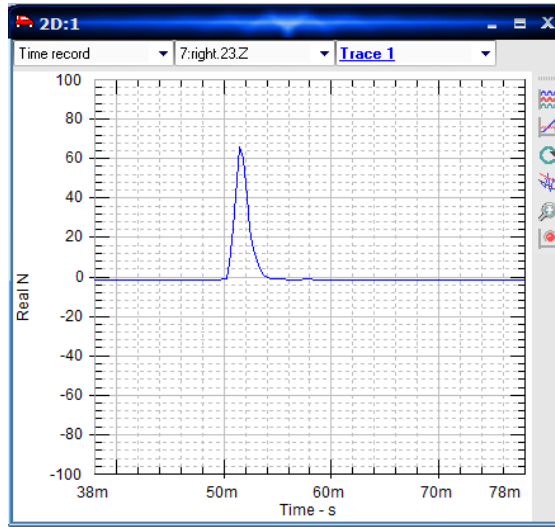


Figure 5.5: Measured impact

The sample frequency has to be set according to the *Shannon-Nyquist theorem* so that no *Aliasing* occurs. Frequencies higher than the *Nyquist frequency* will fold and appear as lower frequencies when the signal is reconstructed. The Shannon-Nyquist theorem states that a signal has to be sampled with a frequency strictly higher than two times the highest measured frequency to avoid aliasing [8]. Expressed as a mathematical inequality according to equation 5.1 where f_s is the sampling frequency and f_{max} is the highest frequency of the investigated signal.

$$f_s > 2f_{max} \quad (5.1)$$

Moreover from equation 5.1, f_{max} can be interpreted as the Nyquist frequency for the case that aliasing should be avoided since the Nyquist frequency is defined as half of the sampling rate [6]. High frequency noise, higher than the Nyquist frequency, will introduce a distortion to the signal. To eliminate the disturbances a low pass filter that rejects frequencies above a certain level is used prior to the sampling with the cut-off at the Nyquist frequency.

Resolution of a discretized signal is determined by the number of bits each sample is assigned, i.e. the bit depth. When sampled a signals momentary value is captured and then quantized. In this process the value is assigned an integer value. The number of possible steps for the sample amplitude is determined by the bit depth. A 16 bit ADC provides $2^{16} = 65536$ steps whereas a 24 bit ADC gives 256 times higher resolution ($2^{24} = 16777216$). The smaller the step size the smaller the round-off error associated with quantization and ultimately a more accurate

description of the continuous signals amplitude.

5.7 Instrument Calibration

Since the purpose of the accelerometer is to capture an acceleration it is of importance that this signal can be translated to a physical quantity that can be relied on. For this reason accelerometers undergoes calibration on regular basis where the sensitivity is determined. Mistreatment of the accelerometer such as a high applied acceleration peak, e.g. from a drop to the floor, will affect the sensor and a calibration has to be carried out.

5.8 Mounting the Accelerometer

There are several ways to mount an accelerometer to a test object and choosing the correct way is critical in order to achieve the correct measured data. What the mounting does is reducing the resonance frequency of the accelerometer, thus reducing the practical frequency range. The mounting introduces additional mass, stiffness and damping to the accelerometer. Five common methods for mounting are presented in table 5.2.

Things needed to be considered when choosing a proper mounting technique for the test are

1. Temperature
2. Frequency span needed
3. Maximum acceleration level
4. Time to set up
5. Repeatability
6. Mounting surface
7. Mass of the test object

The piezoelectric accelerometer is sensitive to temperature fluctuations and the sensitivity change with respect to temperature so when performing a measurement these factors needs to be considered.

Mounting type	Comparable ω_n	Summary
Threaded stud	$31kHz$	Requires holes/threaded holes on the test object
Beeswax	$29kHz$	Sticks to the test object, only up to 40°C , limited acceleration
Cement	$28kHz$	Hard or soft glue for permanent measuring points where stud mounting is not possible
Magnet	$7kHz$	For flat magnetic surfaces, very low resonance frequency
Hand held	$2kHz$	For fast measuring, non-repeatable, the lowest resonance frequency

Table 5.2: Mounting types for accelerometers. Note: The ω_n presented in the second column is for comparison between the types of mounting only [1]

As table 5.2 suggests the mounting affects the resonance frequency of the accelerometer thus reducing the practical measuring range. Knowing the expected frequency range of the test object and taking the first harmonics of the vibrations into consideration is the first step in finding an accelerometer with an appropriate response curve and a fitting mounting technique.

The maximum acceleration level that the accelerometer has to measure does not usually present a big concern, but can introduce problems for sensors mounted with either wax or a magnet. These levels are approximately $100ms^{-1}$ for beeswax and $1000 - 2000ms^{-1}$ for magnetic mounting depending on the mass of the sensor [1]. During certain circumstances machinery have to be shut down to reduce potential personal harm. That can be very expensive so fast mounting can be valuable for that reason.

Finding the measurement point exactly is a prerequisite for repeated measurements thus ensuring validity. The surface has to be flat and smooth where the accelerometer ought to be mounted. It can be necessary to carry out a surface

preparation on the test object prior to testing.

5.8.1 Mass loading effects

The mass of the accelerometer does affect the dynamic properties of the test object and it is desired to have the mass as low as possible. As a general rule of thumb the accelerometer shall be less than 10% of the dynamic mass [2]. When this is not possible to achieve, non-contact methods for measuring should to be consulted.

5.9 Environmental Effects

Due to the principles on which the transducers base their sensing technology, there are undesired effects that has to be taken into consideration in all steps beginning with choosing a suitable accelerometer to actually perform the measurement.

This section will cover the factors from the environment that affects the transducer so that it gives additional outputs that are unrelated to the vibration desired to be measured.

As mentioned earlier piezoelectric materials are affected by temperature, both constant and varying. A high enough temperature will affect the polarization of the sensor. As a certain temperature is reached an artificially polarized sensor, such as PZT, starts to depolarise losing sensitivity as a certain rate permanently. As the temperature increases and reaches the *Curie point* the piezoelectric element is completely destroyed. Besides being rather susceptible to high temperatures the sensitivity of the output depends on temperature, meaning that for an equal load applied on the crystal the output will be different with different ambient temperatures. This has to be taken into consideration when calibrating the accelerometer.

A varying temperature over the piezoelectric element will create an output from the accelerometer. These interferences are of low frequency and therefore constitutes a problem while measuring low frequency vibrations. Efforts has been made to make accelerometers less temperature sensitive by choosing stable natural crystals for the sensing element and having a load design that keeps the sensing element isolated from the accelerometer base. Such a design is described in section 5.3, namely the shear loading design.

Chapter 6

Vibration Testing

6.1 Background to Chapter

This chapter covers resonance search methods and vibration level measurements with focus on experimental modal analysis.

6.2 Introduction

Measuring vibrations are usually interesting from two points of view:

1. Determining the vibration levels and frequency content
2. Determining the dynamic properties of a structure and validate theoretical models

6.3 Experimental Modal Analysis

EMA is a technique used to create an understanding of the dynamic characteristics of a structure by means of experimenting. From response data of a structure the purpose is to extract the modal parameters, namely: natural frequencies, mode shapes and modal damping.

The use of *EMA* extends past the identification of the modal parameters, it also includes correlating and updating Finite Element Method (FEM) models and covers structure modification etcetera. Therefore *EMA* is useful in areas such as design, diagnosis and control [6].

EMA is based on measuring the response of the structure given an input. There are two main options for introducing the input, either with the use of an impact hammer or with the use of an electromechanical shaker. The response, or the output, is measured in the same manner. The impact hammer is practical for field testing or when access is limited and is excellent for quickly testing structures and trouble shooting. It can be used prior to a shaker test to identify the best shaker location.

A shaker however, provides the best results since the input to the structure can be precisely controlled. Any details about a shaker will be omitted in this report although it is concluded that a shaker will provide the best results and therefore will be an interesting topic in the future. This decision is based on the fact that this thesis aims at developing a methodology for vibration testing.

6.3.1 Nodes and Reference Points

Interesting points for impacts on the frame has to be chosen as well as the reference points where the accelerometers are placed. A good nodal representation of the frame is a prerequisite for obtaining good data for analysis. Both with respect to spatial resolution and accessibility.

The nodes should give enough spatial resolution to represent the different modes of interest. The impact location must also allow for repeatable hits, since a number of hits are linearly averaged to ensure good quality measurements. The references points should be placed at points with maximum movement, if the reference instead is placed at a node of zero movement, information for that mode is lost. Stiffness and damping may vary considerably throughout the structure which means that the settings of the acquisition, hardness of the impact hammer tip and the spring back of that part of the structure add up and can produce low quality data.

6.4 Impulsive excitation

For the purpose of conducting resonance search tests there are three main types of tests [3], namely:

1. Impulsive excitation
2. Initial displacement
3. Forced vibrations

Hence by adopting the procedure of EMA into the methodology and rejecting shaker excitation (forced vibration), impulsive excitation was chosen.

For tests where impulsive excitation is used a force of high amplitude and short duration (impulse) is applied to a structure at different locations and both the excitation force and the following response is monitored at one or more locations [3]. This is a so called impact test. The broad frequency band from the impulse excites resonance frequencies in the structure. By monitoring the response between the point of impact and a reference on the structure the transfer function between those points can be obtained. By doing this for multiple locations it is possible to estimate parameters of the structural dynamics of the whole structure [6]. There are several different approaches to an impulsive excitation test regarding the number of inputs and outputs, roving hammer or roving accelerometers.

6.4.1 Impact test

The impact test is performed to collect the data necessary to obtain the transfer functions of a structure. The transfer function is also referred to as the frequency response function (FRF). As discussed in chapter 3, section 3.3 a system can be viewed as a black box. For simple systems as the sdof and tdof, in chapter 3.3, the FRF can be calculated analytically, however this is not possible for complicated structures. Therefore the FRF is instead measured and calculated as the ratio of the Fourier integral transforms of the output and the input between two points on the structure, equation 6.1 [3]

$$H(\omega) = \frac{x(\omega)}{F(\omega)} \quad (6.1)$$

The input force $F(s)$ and the response vibration $x(s)$ are measured with a force transducer and an accelerometer respectively where one location is kept as a reference. The signal is then converted to the frequency domain with fast Fourier transformation. Ideally the FRF depends on the characteristics of the system alone and not on the input. An impact test is carried out as a part of an experimental modal analysis.

6.5 Modal Parameter Estimation

This section briefly describes the process of getting the modal parameters from experimental excitation-response data. For further reading see reference [6] chapter

4. Emphasis will be put on a generalized representation of a transfer function that is important in EMA. The approach differs from the one introduced in chapter 3 section 3.4 but the course of action, how to derive the equation, is similar.

As before the system is described in the general form

$$\mathbf{M}\ddot{\mathbf{x}} + \mathbf{C}\dot{\mathbf{x}} + \mathbf{K}\mathbf{x} = \mathbf{F}$$

Under the assumption of proportional damping a coordinate transformation

$$\mathbf{x} = \mathbf{\Psi}\mathbf{q}$$

can be made [6]

The equation can now be written as

$$\bar{\mathbf{M}}\ddot{\mathbf{q}} + \bar{\mathbf{C}}\dot{\mathbf{q}} + \bar{\mathbf{K}}\mathbf{q} = \mathbf{\Psi}^T\mathbf{F}$$

where

$\mathbf{\Psi}$ is the modal matrix ($n \times n$) of n independent modal vector vectors $[\Psi_1, \Psi_2, \dots, \Psi_n]$

Ψ_i is a column vector

$\bar{\mathbf{M}}$ is the diagonal matrix of modal masses

$\bar{\mathbf{C}}$ is the diagonal matrix of modal damping constants

$\bar{\mathbf{K}}$ is the diagonal matrix of modal stiffness

And assuming the modal vectors are $\mathbf{M}$ -normal

$$M_i = 1$$

$$K_i = \omega_i^2$$

$$C_i = 2\zeta_i\omega_i$$

where

ζ_i is the modal damping ratio.

ω_i is the undamped natural frequency

Omitting the steps prior to the transfer function and displaying the result in equation 6.2.

$$\mathbf{H}(s) = \mathbf{\Psi} \begin{bmatrix} H_1 & & & \\ & H_2 & & \\ & & \ddots & \\ & & & H_n \end{bmatrix} \mathbf{\Psi}^T = [\Psi_1, \Psi_2, \dots, \Psi_n] \begin{Bmatrix} H_1\Psi_1^T \\ H_2\Psi_2^T \\ \vdots \\ H_n\Psi_n^T \end{Bmatrix} = \sum_{r=1}^n H_r\Psi_r\Psi_r^T \quad (6.2)$$

where

$$H_i = \frac{1}{s^2 + 2\zeta_i \omega_i s + \omega_i^2} \text{ for } i = 1, 2, \dots, n$$

Equation 6.2 is a $(n \times n)$ matrix where the transfer characteristics between the response location i and excitation location j is expressed as equation 6.3

$$H_{ij}(s) = \sum_{r=1}^n \frac{(\Psi_i \Psi_j)_r}{[s^2 + 2\zeta_r \omega_r s + \omega_r^2]} \quad (6.3)$$

where

$(\Psi_i)_r$ is the i th element of the r th modal vector

$(\Psi_i \Psi_j)_r$ is the residue (eigenvalue) of the pole, corresponds to the shape of the mode

Equation 6.3 represents the mathematical model of the transfer function on a general form. The modal parameters are extracted through curve fitting the response-excitation data to the model. This process is called model identification, and involves parameter estimation [6].

The steps related to EMA are [6]:

1. Measure an admissible set of excitation-response signals
2. Retrieve the frequency domain transfer function
3. Perform curve fitting of the data to the model
i.e. mode shape, natural frequency and modal damping ratios are extracted
4. System model can be computed

6.6 Test Activities

A separate test plan was made that specifies how the test is intended to be performed. It describes the steps in a way that makes it possible to successfully repeat the test at a later occasion. Following the test plan is the test report. This report documents all relevant aspects of the test and documents any deviations from the test plan. These documents are separate and will not be included in whole in this report. Only partially and in conjunction with the result.

Chapter 7

Result

7.1 Background to chapter

The goal of this thesis is to create a methodology for measuring vibrations on the rotor craft APID60. In this chapter the result produced in this thesis will be stated. Included in the result is test method, equipment, software and data from the vibration analysis.

7.2 Test method

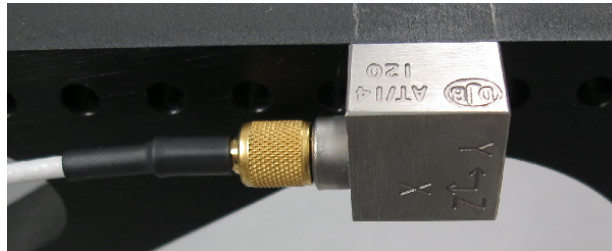
Experimental modal analysis was chosen as part of the methodology to measure vibrations. The method as a whole was aimed at extracting the modal parameters of the test object.

In the EMA the means of exciting frequencies in the structure were accomplished through an impact test with a roving hammer configuration.

For the actual test an external test plan with a test report was made for documenting both the procedure and the result.

7.3 Selected equipment

The equipment chosen are depicted in figure 7.1. For the vibration measurements the accelerometer AT/14, figure 7.1a, was used along with the impact hammer Dytran 5850A, figure 7.1b. The measured data was handled by the VibPilot front end, figure 7.1c, and analyzed in the software m+p SO Analyzer, figure 7.1d.



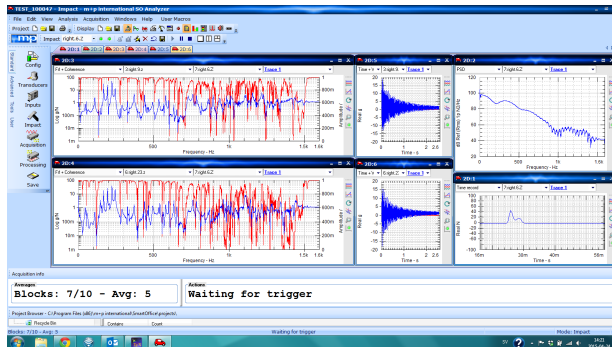
(a) AT/14



(b) Dytran 5850A



(c) The 8-channel VibPilot



(d) m+p SO Analyzer

Figure 7.1: Equipment and software chosen for the data acquisition, testing and analysis.

Type	Description	Specifications
Accelerometer	DJB Instruments AT/14	• Weight: 13 gram • Sensitivity: 100mV/g • Frequency response: 7000 Hz • Maximum g limit: 450g • IEPE • Piezoelectric • Number of axis: 3
Impact hammer	Dytran 5850A	• Head weight: 150 gram • Interchangeable impact tip • IEPE
Data acquisition	VibPilot	• 8-channels with BNC connectors • 102.4 kHz simultaneous sampling • ICP sensor • AC/DC supply 20 W power consumption
Software	m+p SO Analyzer	• Impact test data acquisition • Natural frequency and mode shape estimation

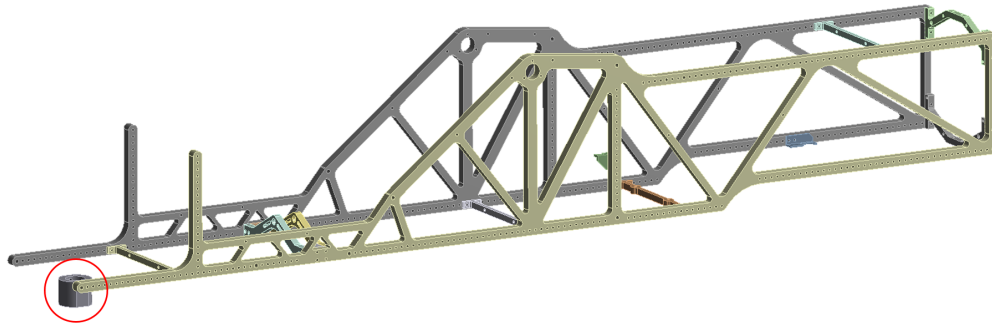
Table 7.1: This table lists the equipment and software chosen for data acquisition and analysis along with some of the specifications.

7.4 Experimental modal analysis

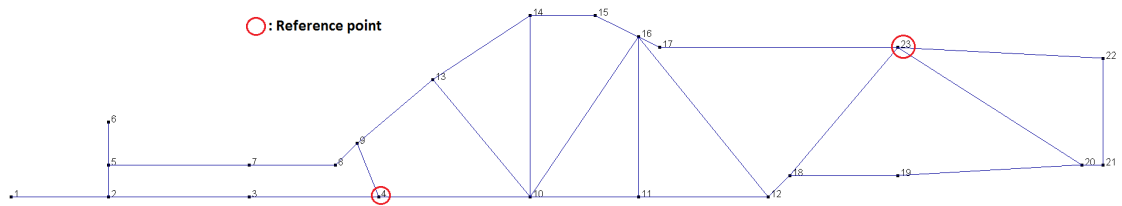
Prior to the impact test 23 nodes were chosen as impact points where two of them acted as references points for the accelerometers, see figure 7.2b.

As part of the comparative study a point mass was securely tightened to the frame prior to the second impact test to alter the modal parameters of the test object. The weight was measured to 406.8 grams which for the test object is significant from a mass loading point of view. However, the shape was arbitrarily chosen, see figure 7.2a.

There are several methods available for parameter estimation, the one chosen in this thesis was the *Multi-Degree of Freedom Curve Fitting*. Computationally more demanding but offers better accuracy and distinction between closely spaced modes [6].



(a) Test object with point mass. Location of point mass is marked with a circle.



(b) Nodal representation of the frame with accelerometer reference points marked.

Figure 7.2: Representation of test object in CAD environment and m+p SO Analyzer.

7.5 Results from EMA

7.5.1 Validation

Results from EMA	
Frequency [Hz]	Modal damping [%]
146,46	0,417
156,10	0,296
168,82	0,084
194,28	0,046
205,94	0,552
212,17	0,384
223,68	0,113
245,58	0,133
277,43	0,147
289,87	0,633

Table 7.2: Results of experimental modal analysis of test object.

Results from EMA	
Frequency [Hz]	Modal damping [%]
135,92	0,409
144,00	0,497
163,61	0,083
172,29	0,249
194,35	0,079
200,10	0,547
211,65	0,319
213,91	0,399
241,96	0,233
266,44	0,291
289,73	0,663

Table 7.3: Results of experimental modal analysis of test object with point mass added.

Results from ANSYS
Frequency [Hz]
147,34
156,89
169,96
196,71
210,87
215,14
225,90
250,31
276,52
291,60

Table 7.4: Results from modal analysis of test object in ANSYS.

Results from ANSYS
Frequency [Hz]
138,97
141,39
165,09
174,60
197,18
201,56
215,21
218,49
243,49
268,25
291,34

Table 7.5: Results from modal analysis of test object in ANSYS with point mass added.

Chapter 8

Conclusion

The work of this thesis was aimed at deriving a methodology for measuring vibrations on the VTOL RPAS's UAV APID60. Through the study, based on the results from the EMA compared with the numerical modal analysis, it is concluded that the use of this type vibration measurements and analysis included in this method will be a valuable contribution to the future work related to developing the aircraft in both design and evaluation.

Given external demands on APID60 needed for either certification or operation in certain areas there are vibration related demands on the structure of the APID60. These demands are translated to constraints on the dynamic properties of the structure which are found with EMA and used to validate theoretical models.

If enough effort is put in EMA computer models of the helicopter can be validated. This means that accurately predicting the effects in frequencies and modes when adding or removing e.g. different payloads can be done in a computer environment. Time and cost of changes can therefore be kept lower while quality can be maintained or even increased.

Not only is vibration measurements important for developing the helicopter but this thesis also shows that know-how in this field of engineering is required in the process of getting a certification of the product.

It is also shown that equipment for measuring vibrations are costly and requires maintaining in form of calibration, so building a good relation and cooperation with companies providing the equipment will be an efficient way how to introduce these kind of tests into the business, since building a proper base of measuring equipment is costly.

As can be seen from the comparison in table 8.1 and 8.2 the frequencies for

each mode resides slightly lower for all except one mode compared to the computer simulation performed in ANSYS. This is as it should be since the model used in ANSYS excluded fasteners and thus had a lower mass and did not take friction between fasteners etc. into account.

Comparison		
EMA	ANSYS	Difference
Frequency [Hz]	Frequency [Hz] [%]	[Δ Hz]
146,46	147,34	-0,88
156,10	156,89	-0,79
168,81	169,96	-1,15
194,29	196,71	-2,42
205,94	210,87	-4,93
212,17	215,14	-2,97
223,68	225,90	-2,22
245,58	250,31	-4,73
277,43	276,52	0,91
289,87	291,60	-1,73

Table 8.1: Results for corresponding modes compared between the experimental modal analysis and results from ANSYS.

Comparison		
EMA	ANSYS	Difference
Frequency [Hz]	Frequency [Hz]	[Δ Hz]
135,92	138,97	-3,05
144,00	141,39	2,61
163,61	165,09	-1,48
172,29	174,60	-2,31
194,35	197,18	-2,83
200,10	201,56	-1,46
211,65	215,21	-3,56
213,91	218,49	-4,58
241,96	243,49	-1,53
266,44	268,25	-1,81
289,73	291,34	-1,61

Table 8.2: Results for corresponding modes compared between the experimental modal analysis and results from ANSYS for the case with point mass added.

Chapter 9

Discussion

During the work of this thesis many crossroads had to be faced. One of them was either go for a shaker or an impact hammer. The choice obviously fell on the impact hammer although a shaker provides the best result since the input to the structure can be precisely controlled. The reason why the choice fell on the impact hammer was due to time and money constraints. By choosing the impact hammer and evaluating the results conclusions about the impact hammer being adequate or not could be made. This can support future decisions in vibration analysis when standing at a similar crossroad.

A soft hammer tip was used during the impact test so proper excitation was only acquired for frequencies up to roughly $450Hz$, see figure 9.1, and excitation input was unstable up to $130Hz$, see figure 9.2. Therefore only acceptable results could be extracted in the frequency range from $130Hz$ to $300Hz$. This does constitute a problem when modes in a certain frequency range are to be extracted. A possible way of overcoming this is to perform the EMA with a shaker instead of a hammer.

During the impact test the frame was supported vertically by four damping feet. These damping feet reduced the motion of the horizontal impact at the area in close proximity to the support. To what extent this affects the result of the dynamic properties of the structure is hard to tell. Mode shapes and resonance frequency could be affected as well as the vibration decay rate. Trying different materials with altered stiffness and damping chosen specifically for the weight and shape of the test object should be considered. Another way of support could be accomplished by letting the frame hang from the ceiling, supported by elastic strings.

Only used one axis of measurement in the accelerometer, so a one axial accelerometer would be sufficient for this test. Though, the accelerometer was chosen in the beginning when the test plan yet did not exist.

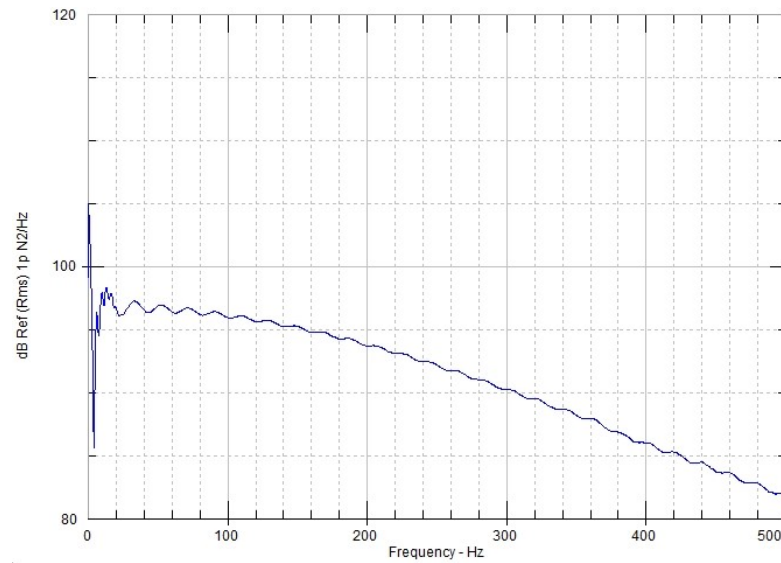


Figure 9.1: The power spectral density of a general impact on the test object with a soft hammer tip. Here the impact experiences a decrease of roughly 15dB up to 450Hz

The process of choosing accelerometers and a corresponding logger proved to be more complicated than first thought. This had several reasons. Firstly, getting the pricing information and secondly, containing it within the budget of the thesis work. Prices for the equipment is rarely written on the web page of the manufacturer. Instead the prices of the equipment is accessible through one of the manufacturers distributors, who has to be contacted either by email or phone, or simply by a visit.

Going through accelerometers and data loggers this way takes time. Appropriate accelerometers and loggers have many times been discarded as an option due to their hefty price tags, where just one accelerometer could be well above the budget of this thesis and loggers that was multiples of the price of an accelerometer.

The resonance search test for determining the resonance frequencies is to be carried out before a main test where a shaker is used, so this work makes up the foundation for further testing of the APID60. Of course, for the work to be of use, it has to be documented properly. This means that the test plan, and test procedure documents will be the key for following testing. Let say if the documentation of the testing is inadequate a new resonance search test has to be performed. If

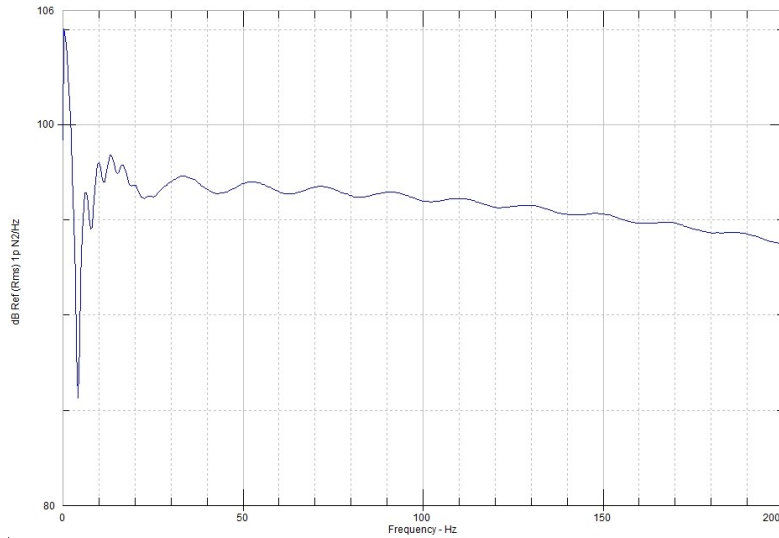


Figure 9.2: The power spectral density of a general impact. Note the unstable energy input in the low frequency region, below 50Hz .

so is the case, then at least the results from the tests performed in this thesis can be used as a reference to some extent.

The extensive literature study of this thesis lead to a few realistic realisations of permanent vibration monitoring and performable changes of the helicopter to enhance the life time of certain parts from a vibrational perspective. Permanent vibration monitoring is widely used in the industry to look for indications of failure and to alarm when certain vibration levels are reached. One can easily understand the benefits of having a permanent monitoring system on a autonomously flying vehicle such as the APID60.

The idea of the permanent monitoring system is to replace the tactile and acoustical perception of the pilot. When a pilot experience increased levels of vibrations, structural or not, a safe landing should be performed.

Almost everything in a helicopter is flight critical, so foreseeing or predicting failures of these critical components is an important step in increasing quality, reliability and safety of the UAS. With a permanent monitoring system these increases of vibration levels can be distinguished and send to the ground control station where the operator can take action. The helicopter can also be programmed to land on a pre-determined spot after the alarm. This might both save the helicopter and prevent third part damage. Such a system can be integrated in the frame on strategical places to grant higher environmental protection of the sensors. The use

of these sensors are applicable in the development process of the helicopter too. This requires though that the sensor can be purchased cheaply since they add to the total cost of the helicopter. Luckily MEMS accelerometers are getting cheaper and cheaper even as the performance of them increase. Before optimizing the performance vs cost with respect of their task a thorough test with high quality piezoelectric accelerometers can be conducted for reference data of the different cases.

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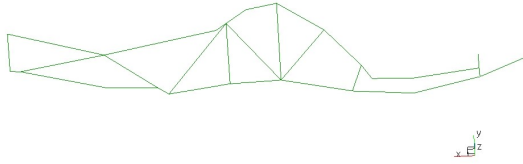
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Appendices

Appendix A

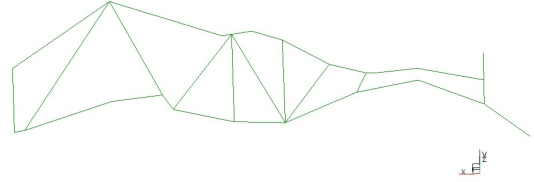
Modes from EMA

right
f = 146,46 Hz D = 0,41 % EMA-MDOF



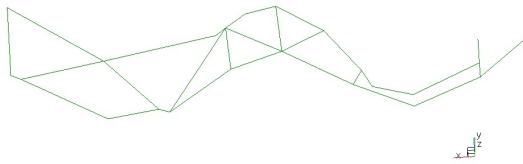
(a) 146,46 Hz

right
f = 146,46 Hz D = 0,41 % EMA-MDOF



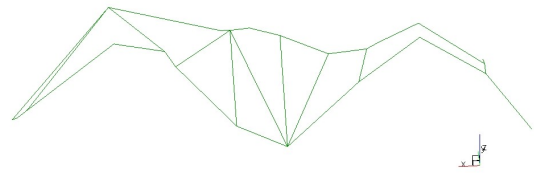
(b) 146,46 Hz

right
f = 156,10 Hz D = 0,29 % EMA-MDOF



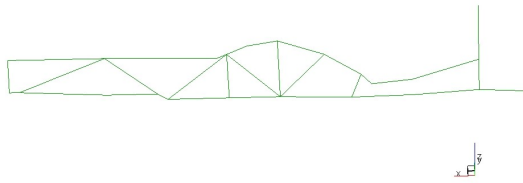
(c) 156,10 Hz

right
f = 156,10 Hz D = 0,29 % EMA-MDOF



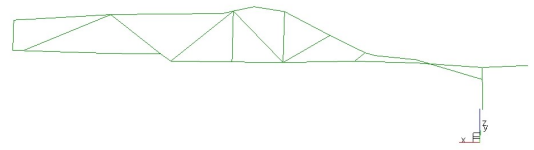
(d) 156,10 Hz

right
f = 168,82 Hz D = 0,08 % EMA-MDOF



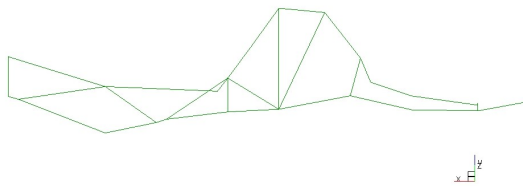
(e) 168,82 Hz

right
f = 168,82 Hz D = 0,08 % EMA-MDOF



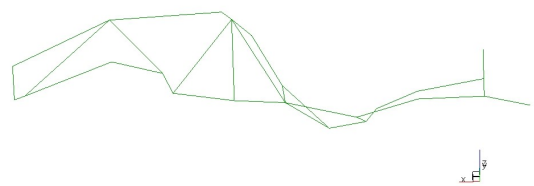
(f) 168,82 Hz

right
f = 194,28 Hz D = 0,05 % EMA-MDOF



(g) 194,28 Hz

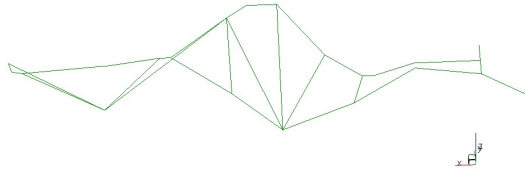
right
f = 194,28 Hz D = 0,05 % EMA-MDOF



(h) 194,28 Hz

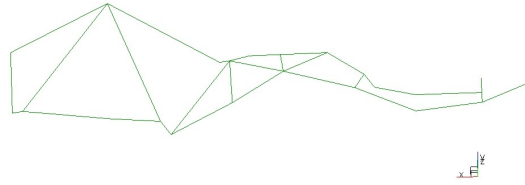
Figure A.1: Still images of the mode shapes from the experimental modal analysis. No point mass added.

right
f = 205,94 Hz D = 0,53 % EMA-MDOF



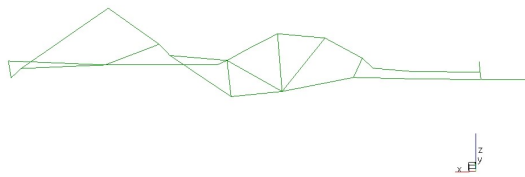
(a) 205,94 Hz

right
f = 205,94 Hz D = 0,53 % EMA-MDOF



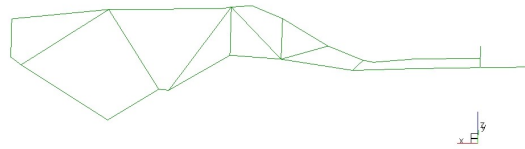
(b) 205,94 Hz

right
f = 212,17 Hz D = 0,39 % EMA-MDOF



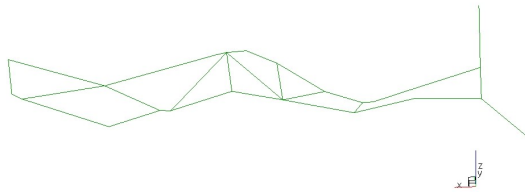
(c) 212,17 Hz

right
f = 212,17 Hz D = 0,39 % EMA-MDOF



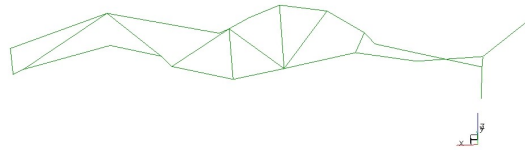
(d) 212,17 Hz

right
f = 223,68 Hz D = 0,11 % EMA-MDOF



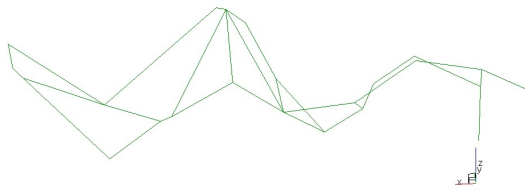
(e) 223,68 Hz

right
f = 223,68 Hz D = 0,11 % EMA-MDOF



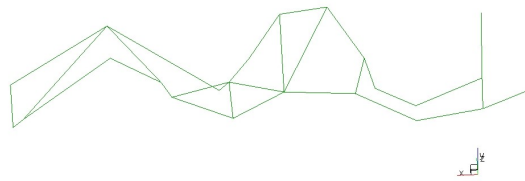
(f) 223,68 Hz

right
f = 245,58 Hz D = 0,25 % EMA-MDOF



(g) 245,58 Hz

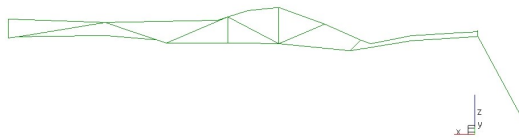
right
f = 245,58 Hz D = 0,25 % EMA-MDOF



(h) 245,58 Hz

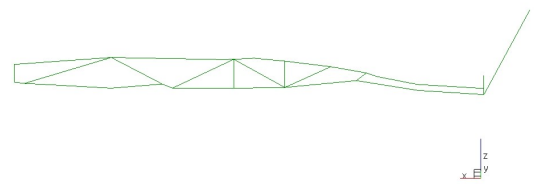
Figure A.2: Still images of the mode shapes from the experimental modal analysis. No point mass added.

right
f = 277,43 Hz D = 0,15 % EMA-MDOF



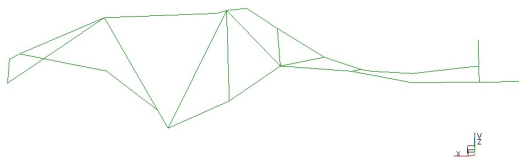
(a) 277,43 Hz

right
f = 277,43 Hz D = 0,15 % EMA-MDOF



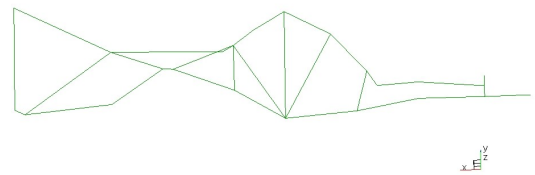
(b) 277,43 Hz

right
f = 289,87 Hz D = 0,63 % EMA-MDOF



(c) 289,87 Hz

right
f = 289,87 Hz D = 0,63 % EMA-MDOF



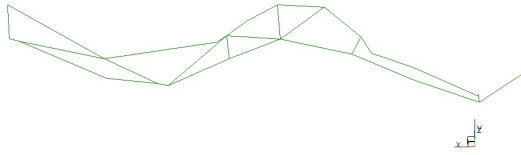
(d) 289,87 Hz

Figure A.3: Still images of the mode shapes from the experimental modal analysis. No point mass added.

Appendix B

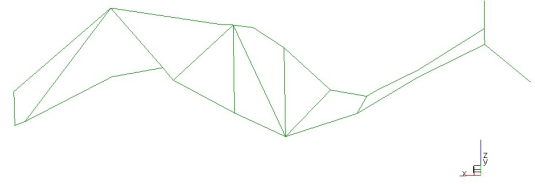
Modes from EMA - Point mass

right
f = 135,92 Hz D = 0,41 % EMA-MDOF



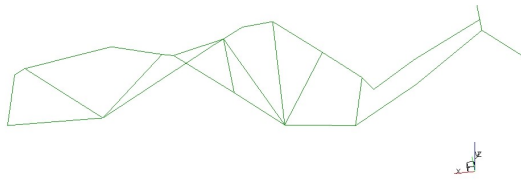
(a) 135,92 Hz

right
f = 135,92 Hz D = 0,41 % EMA-MDOF



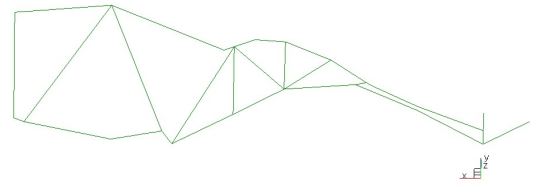
(b) 135,92 Hz

right
f = 144,00 Hz D = 0,50 % EMA-MDOF



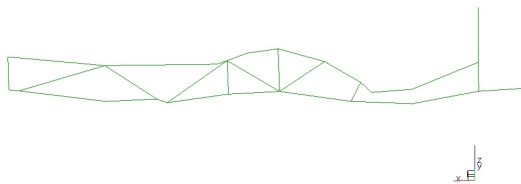
(c) 144,00 Hz

right
f = 144,00 Hz D = 0,50 % EMA-MDOF



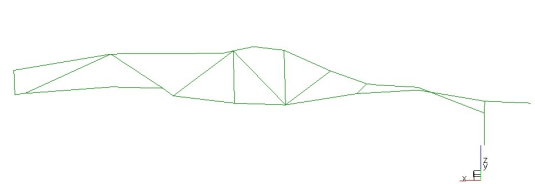
(d) 144,00 Hz

right
f = 163,61 Hz D = 0,08 % EMA-MDOF



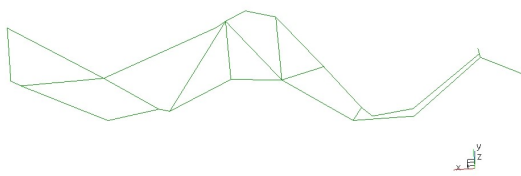
(e) 163,61

right
f = 163,61 Hz D = 0,08 % EMA-MDOF



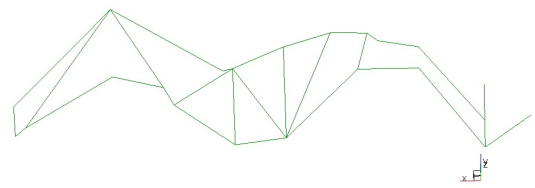
(f) 163,61 Hz

right
f = 172,29 Hz D = 0,25 % EMA-MDOF



(g) 172,29 Hz

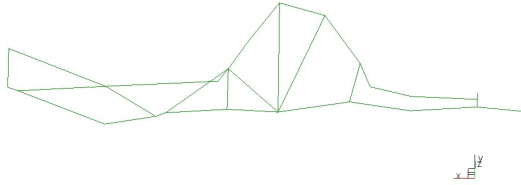
right
f = 172,29 Hz D = 0,25 % EMA-MDOF



(h) 172,29 Hz

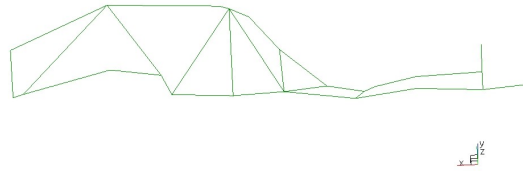
Figure B.1: Still images of the mode shapes from the experimental modal analysis.

right
f = 194,35 Hz D = 0,08 % EMA-MDOF



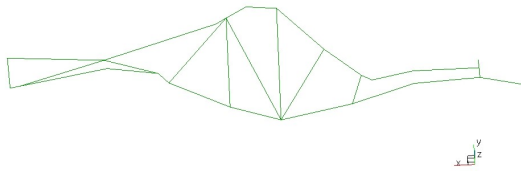
(a) 194,35 Hz

right
f = 194,35 Hz D = 0,08 % EMA-MDOF



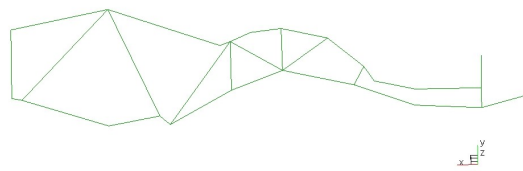
(b) 194,35 Hz

right
f = 200,10 Hz D = 0,55 % EMA-MDOF



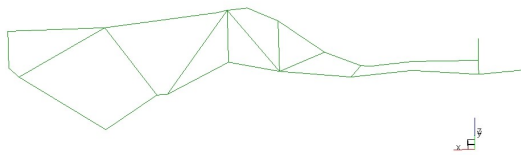
(c) 200,10 Hz

right
f = 200,10 Hz D = 0,55 % EMA-MDOF



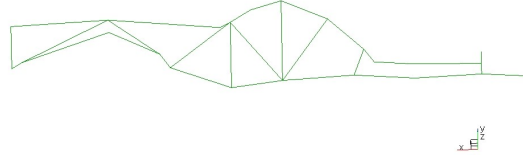
(d) 200,10 Hz

right
f = 211,65 Hz D = 0,32 % EMA-MDOF



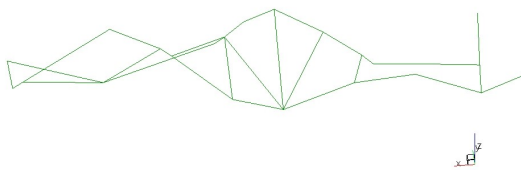
(e) 211,65 Hz

right
f = 211,65 Hz D = 0,32 % EMA-MDOF



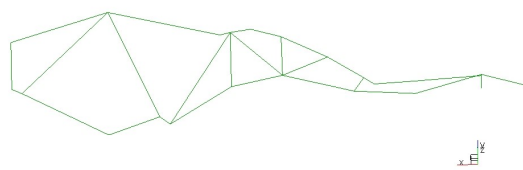
(f) 211,65 Hz

right
f = 213,91 Hz D = 0,40 % EMA-MDOF



(g) 213,91 Hz

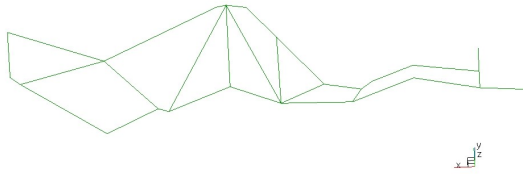
right
f = 213,91 Hz D = 0,40 % EMA-MDOF



(h) 213,91 Hz

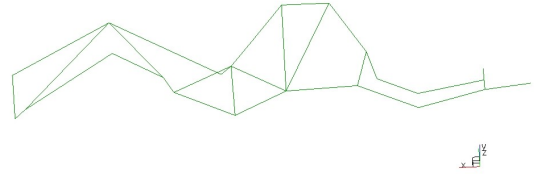
Figure B.2: Still images of the mode shapes from the experimental modal analysis.

right
f = 241,96 Hz D = 0,23 % EMA-MDOF



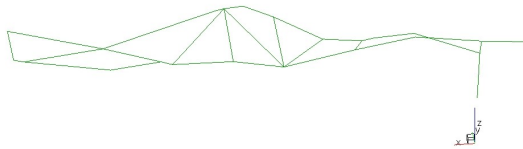
(a) 241,96 Hz

right
f = 241,96 Hz D = 0,23 % EMA-MDOF



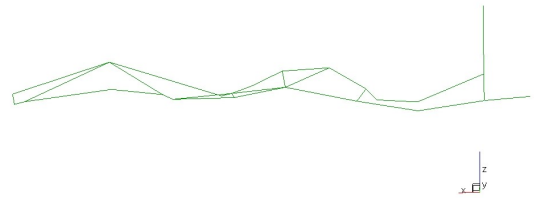
(b) 241,96 Hz

right
f = 266,44 Hz D = 0,29 % EMA-MDOF



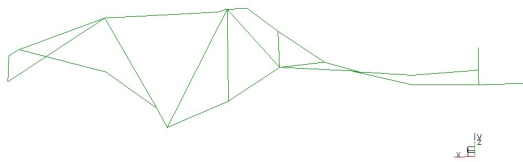
(c) 266,44 Hz

right
f = 266,44 Hz D = 0,29 % EMA-MDOF



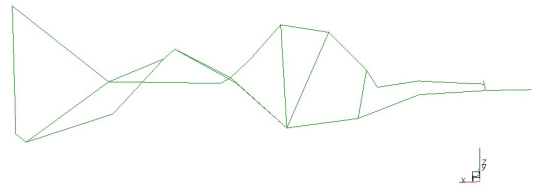
(d) 266,44 Hz

right
f = 289,73 Hz D = 0,66 % EMA-MDOF



(e) 289,73 Hz

right
f = 289,73 Hz D = 0,66 % EMA-MDOF

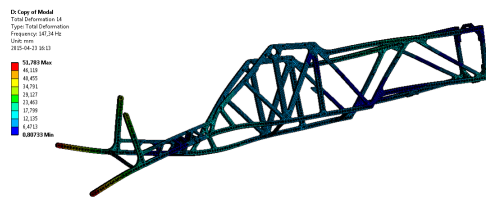


(f) 289,73 Hz

Figure B.3: Still images of the mode shapes from the experimental modal analysis.

Appendix C

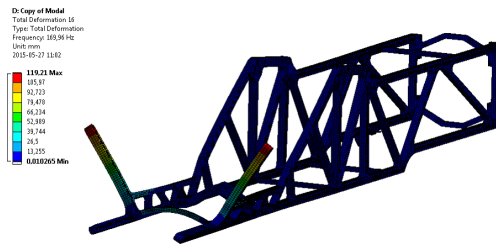
Modes from ANSYS



(a) 147,34 Hz



(b) 156,89 Hz



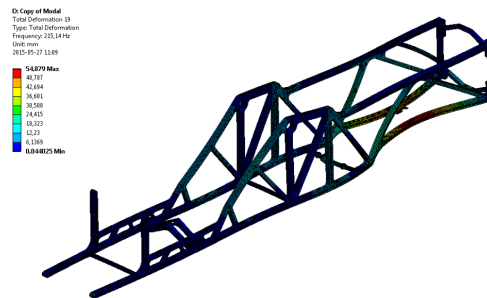
(c) 169,96 Hz



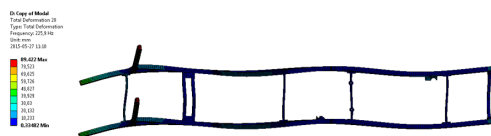
(d) 196,71 Hz



(e) 210,87 Hz



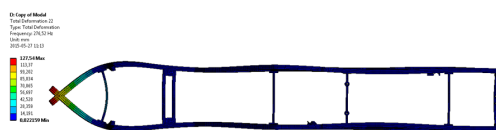
(f) 215,14 Hz



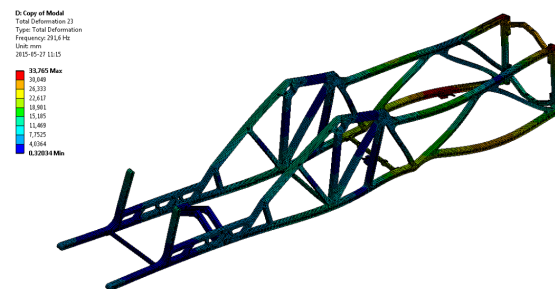
(g) 225,90 Hz



(h) 250,31 Hz



(i) 276,52 Hz



(j) 291,60 Hz

Figure C.1: Still images of the mode shapes from ANSYS.

Appendix D

Modes from *ANYSY* - Point mass

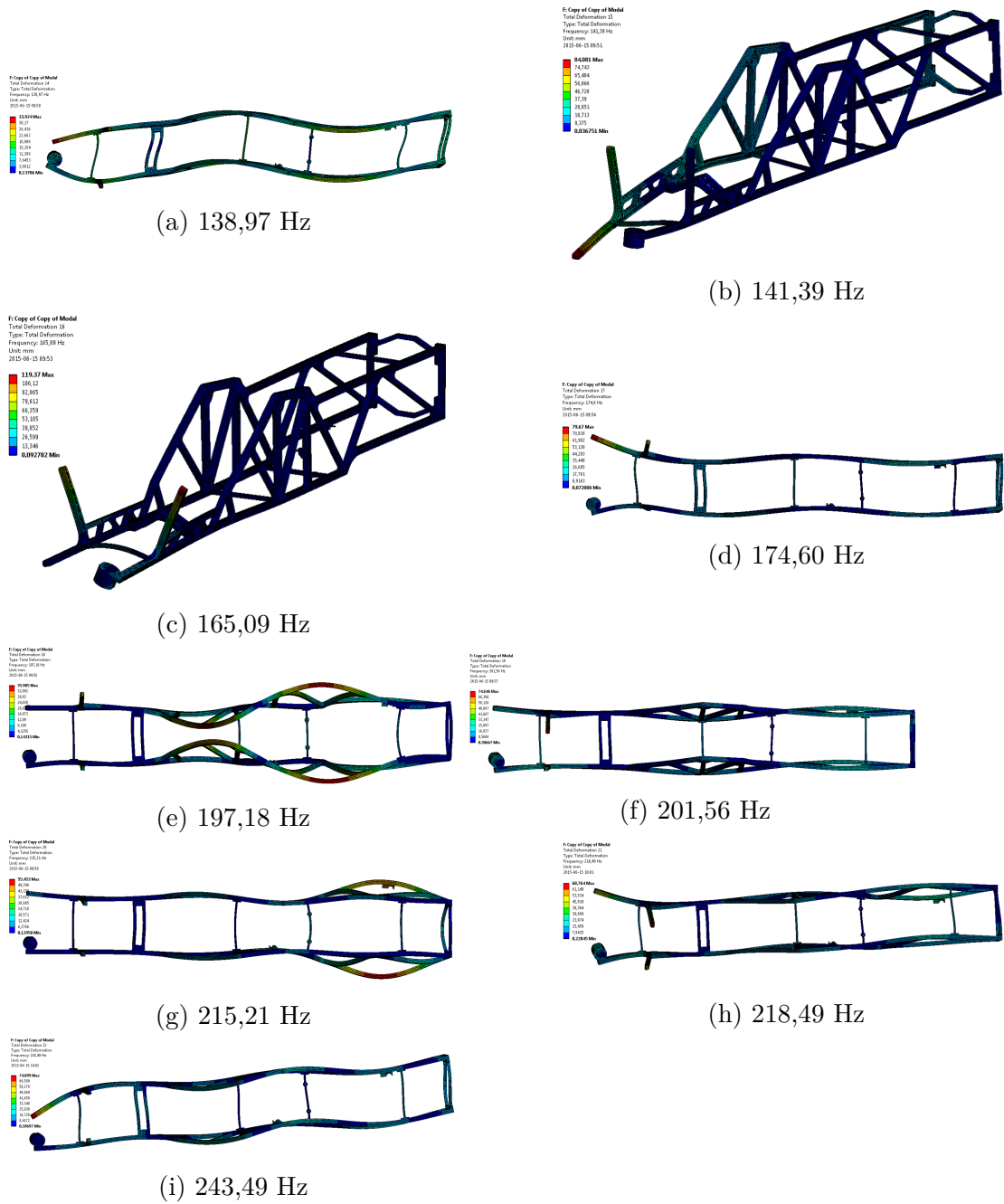
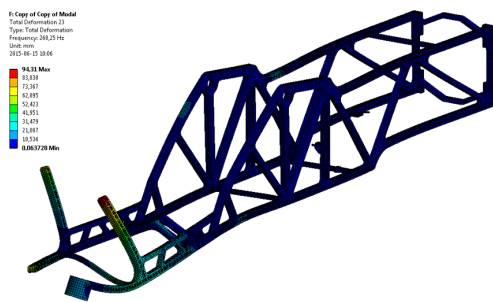
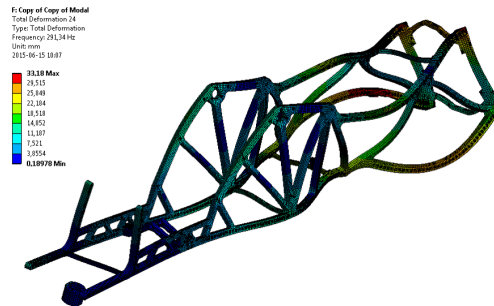


Figure D.1: Still images of the mode shapes from ANSYS with point mass.



(a) 268,25 Hz



(b) 291,34 Hz

Figure D.2: Still images of the mode shapes from ANSYS with point mass.

Appendix E

Estimation of Dynamic Mass

$$\frac{m_{acc}}{m_{dyn}} = 0.1 \quad (E.1)$$

according to [2] where

m_{acc} is the mass of the accelerometer [kg]

$$m_{dyn} = Al\rho \quad (E.2)$$

where

A is the cross section area [m^2]

ρ is the density of aluminum [kg/m^3]

l is the total length of the section

This can be translated into a total length of the dynamic mass, this should be the minimal sectional length that moves. The M4 holes on the frame are neglected.

$$l \geq \frac{10m_{acc}}{A\rho} \quad (E.3)$$

with

$$m_{acc} = 0.013kg$$

$$A = 0.0225m^2$$

$$\rho = 2700kg/m^3$$

$$l = 21.4cm$$

Given the location of the accelerometers the total length of the dynamic mass is larger than $l = 21.4cm$ for both accelerometers.